

BODY

REFER TO 1990 CELICA REPAIR MANUAL (Pub. No. RM149U)

NOTE: The following pages contain only the points which differ from the above listed manual.

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BODY DIMENSIONS	BO-2

BODY DIMENSIONS

General Information

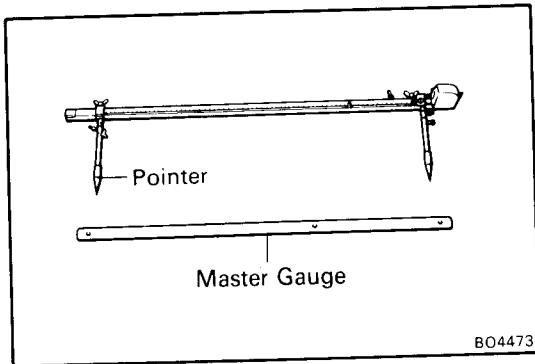
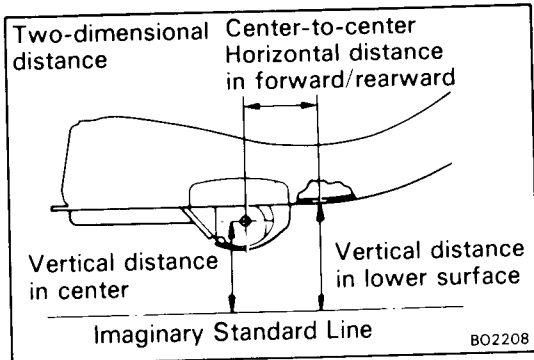
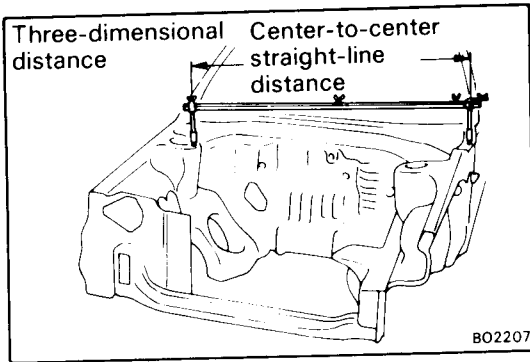
1. BASIC DIMENSIONS

- (a) There are two types of dimensions in the diagram.
(Three-dimensional distance)

- Straight-line distance between the centers of two measuring points.

(Two-dimensional distance)

- Horizontal distance in forward/rearward between the centers of two measuring points.
 - The height from an imaginary standard line.
- (b) Incases in which only one dimension is given, left and right are symmetrical.
- (c) The dimensions in the following drawing indicate actual distance. Therefore, please use the dimensions as a reference.

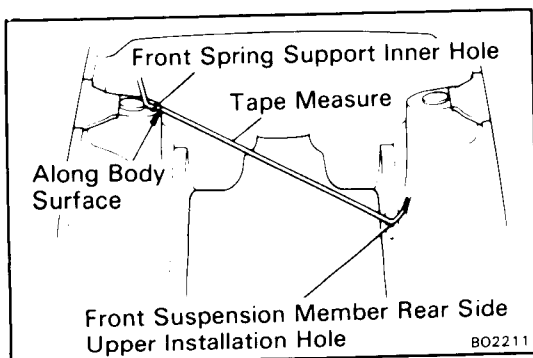
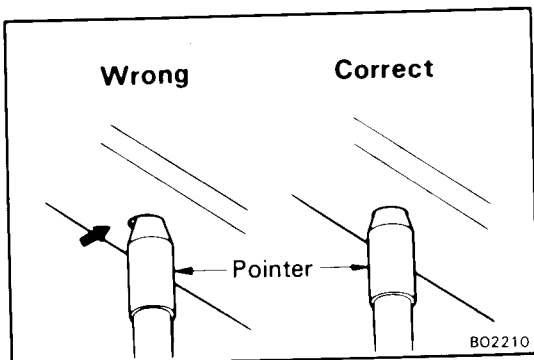


2. MEASURING

- (a) Basically, all measurements are to be done with a tracking gauge. For portions where it is not possible to use a tracking gauge, a tape measure should be used.
- (b) Use only a tracking gauge that has no looseness in the body, measuring plate, or pointers.

HINT:

1. The height of the left and right pointers must be equal.
2. Always calibrate the tracking gauge before measuring or after adjusting the pointer height.
3. Take care not to drip the tracking gauge or otherwise shock it.
4. Confirm that the pointers are securely in the holes.



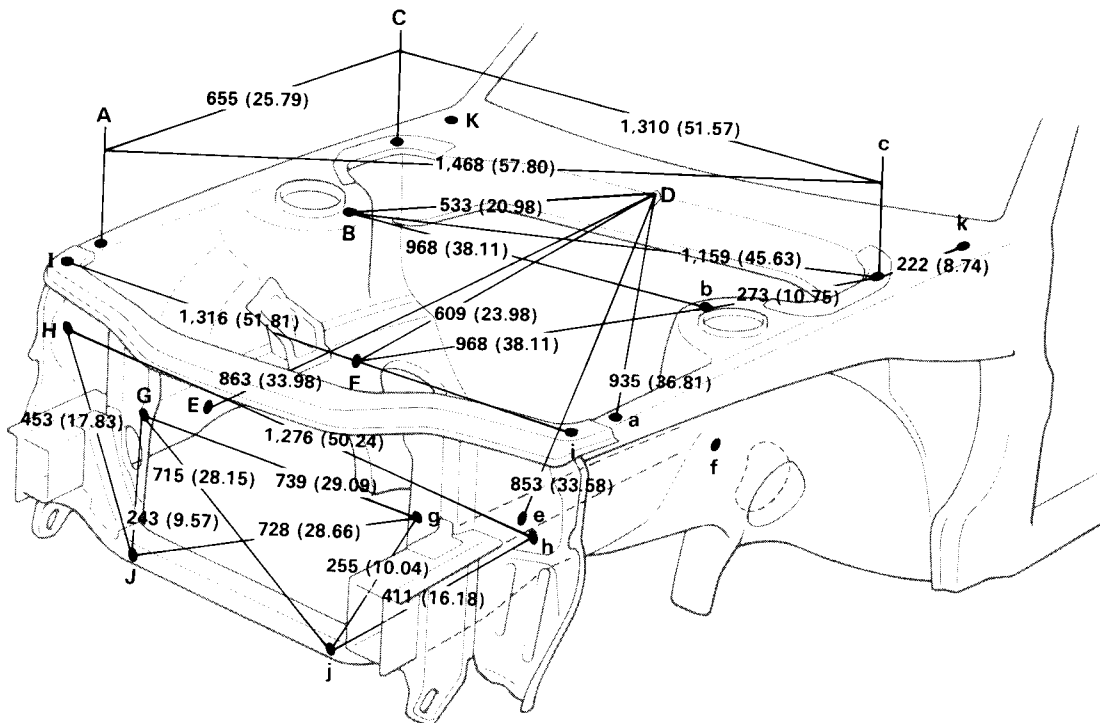
- (c) When using a tape measure, avoid twists and bends in the tape.

- (d) When tracking a diagonal measurement from the front spring support inner hole to the suspension member upper rear installation hole, measure along the front spring support panel surface.

Body Dimensions

ENGINE COMPERTMENT

(Three-Dimensional Distance)

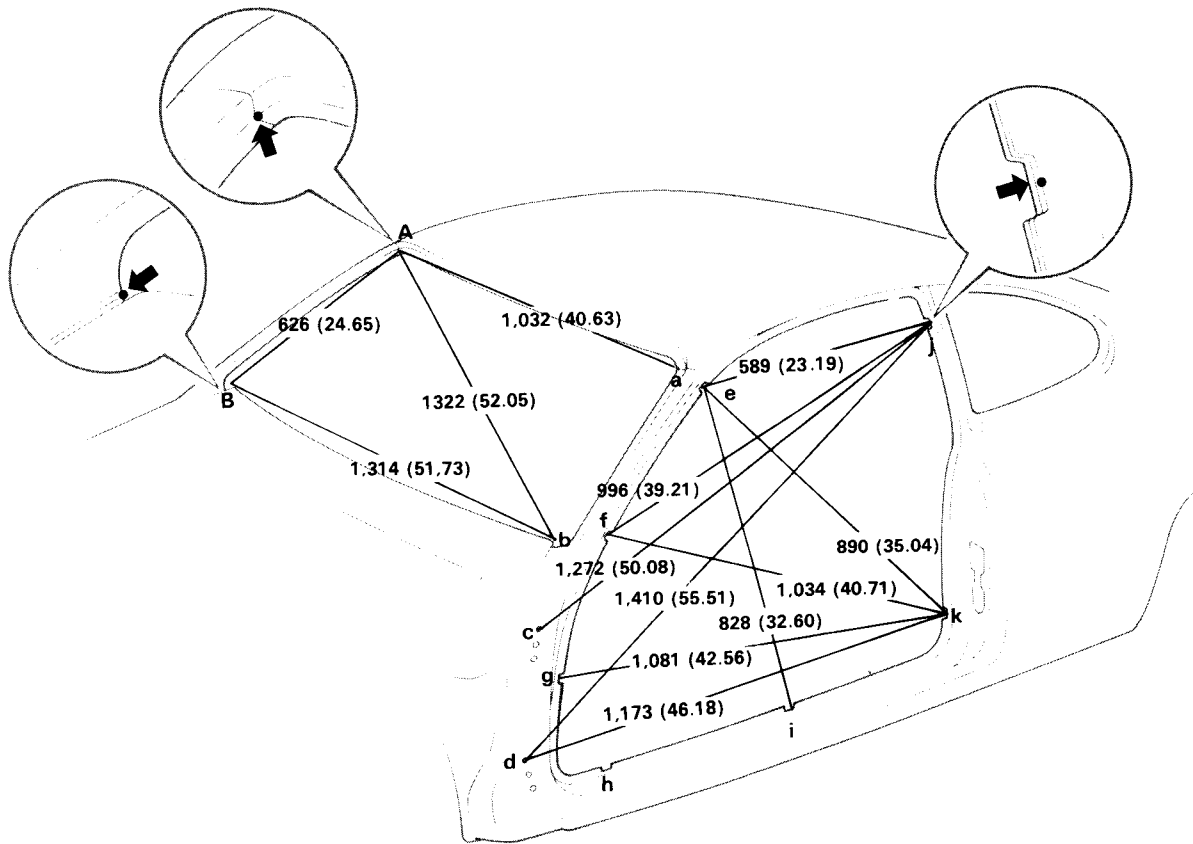


A – I or a – i	C – I or c – i	C – i or c – I	I – i
140 (5.51)	794 (31.26)	1,519 (59.80)	1,281 (50.43)

Symbol	Name	Hole dia.
A,a	Front fender apron standard hole	10 (0.39)
B,b	Cowl top to apron brace standard hole	11 (0.43)
C,c	Cowl top to apron brace standard hole	10 (0.39)
D	Cowl top panel center mark	–
E,e	Front side member standard hole	15 (0.59)
F,f	Front side member standard hole	15 (0.59)
G,g	Radiator support standard hole	10 (0.39)
H,h	Radiator support standard hole	10 (0.39)
I,i	Front fender installation nut	6 (0.24) nut
J,j	Cooler condenser installation nut	6 (0.24) nut
K,k	Cowl top panel standard hole	10 (0.39)

BODY OPENING AREAS (Side View)

(Three-Dimensional Distance)



Vehicle Dimensions Left ↔ Right

C – c	D – d	E – e	F – f	G – g	H – h	I – i	J – j	K – k
1,495 (58.86)	1,527 (60.12)	1,086 (42.76)	1,308 (51.50)	1,373 (54.06)	1,450 (57.09)	1,450 (57.09)	1,118 (44.02)	1,433 (56.42)

E – i or e – I	E – j or e – J	E – k or e – K	F – h or f – H	F – j or f – J	F – k or f – K	G – j or g – J	G – k or g – K	I – j or i – J
1,504 (59.21)	1,249 (49.17)	1,532 (60.31)	1,494 (58.82)	1,567 (61.69)	1,715 (67.52)	1,731 (68.15)	1,771 (69.72)	1,590 (62.60)

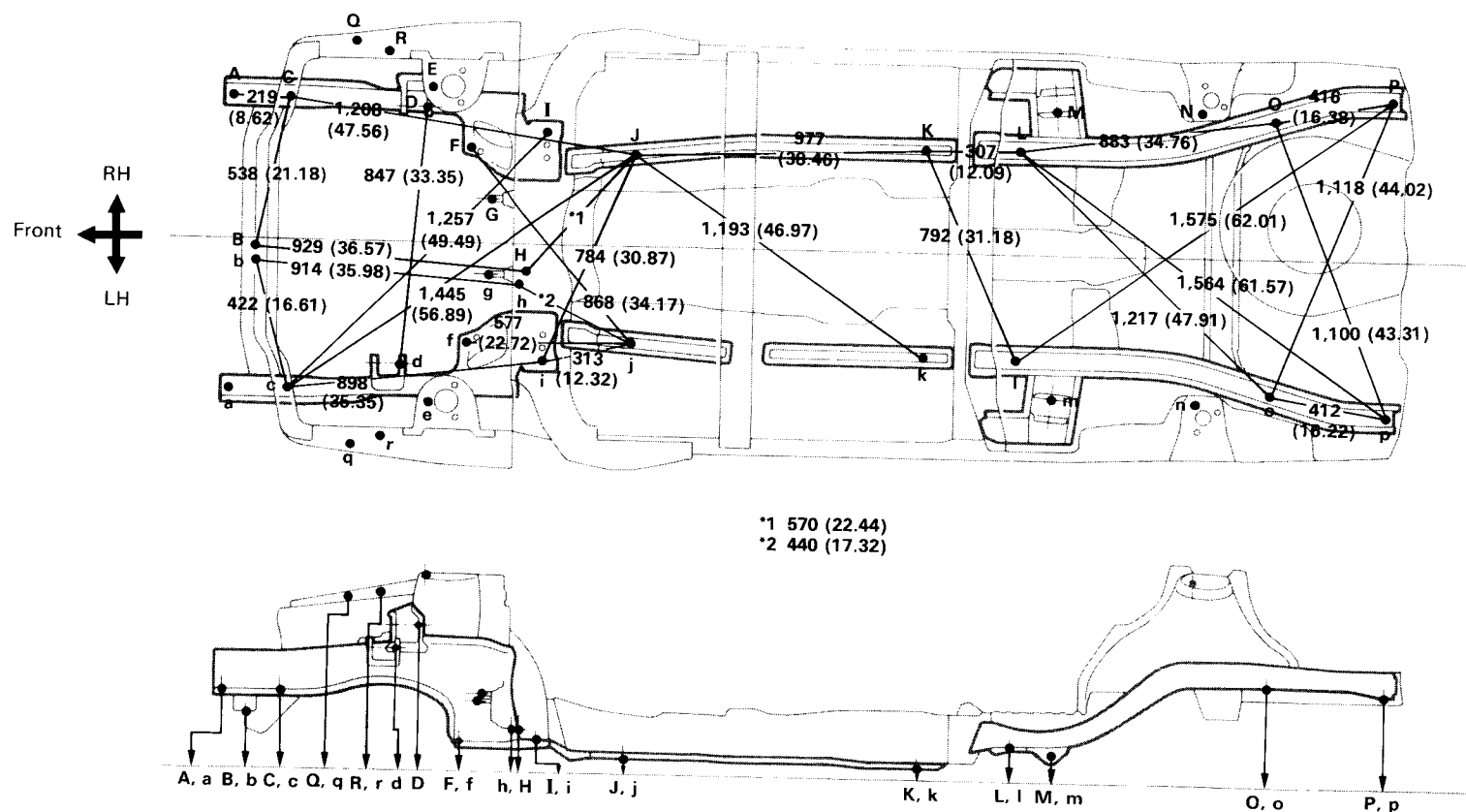
HINT: For Symbols, Capital letters indicate right side of vehicle, small letters indicate left side of vehicle. (Seen from rear)

mm (in.)

Symbol	Name	Hole dia.
A,a	Roof panel/Front body pillar adjoining portion	–
B,b	Front body pillar/Cowl panel adjoining portion	–
C,c	Front door hinge installation nut	8 (0.31) nut
D,d	Front door hinge installation nut	8 (0.31) nut
E,e	Front body pillar assembly mark	–
F,f	Front body pillar assembly mark	–
G,g	Front body pillar assembly mark	–
H,h	Rocker panel assembly mark	–
I,i	Rocker panel assembly mark	–
J,j	Quarter panel assembly mark	–
K,k	Quarter panel assembly mark	–

UNDER BODY

(Three-Dimensional Distance)

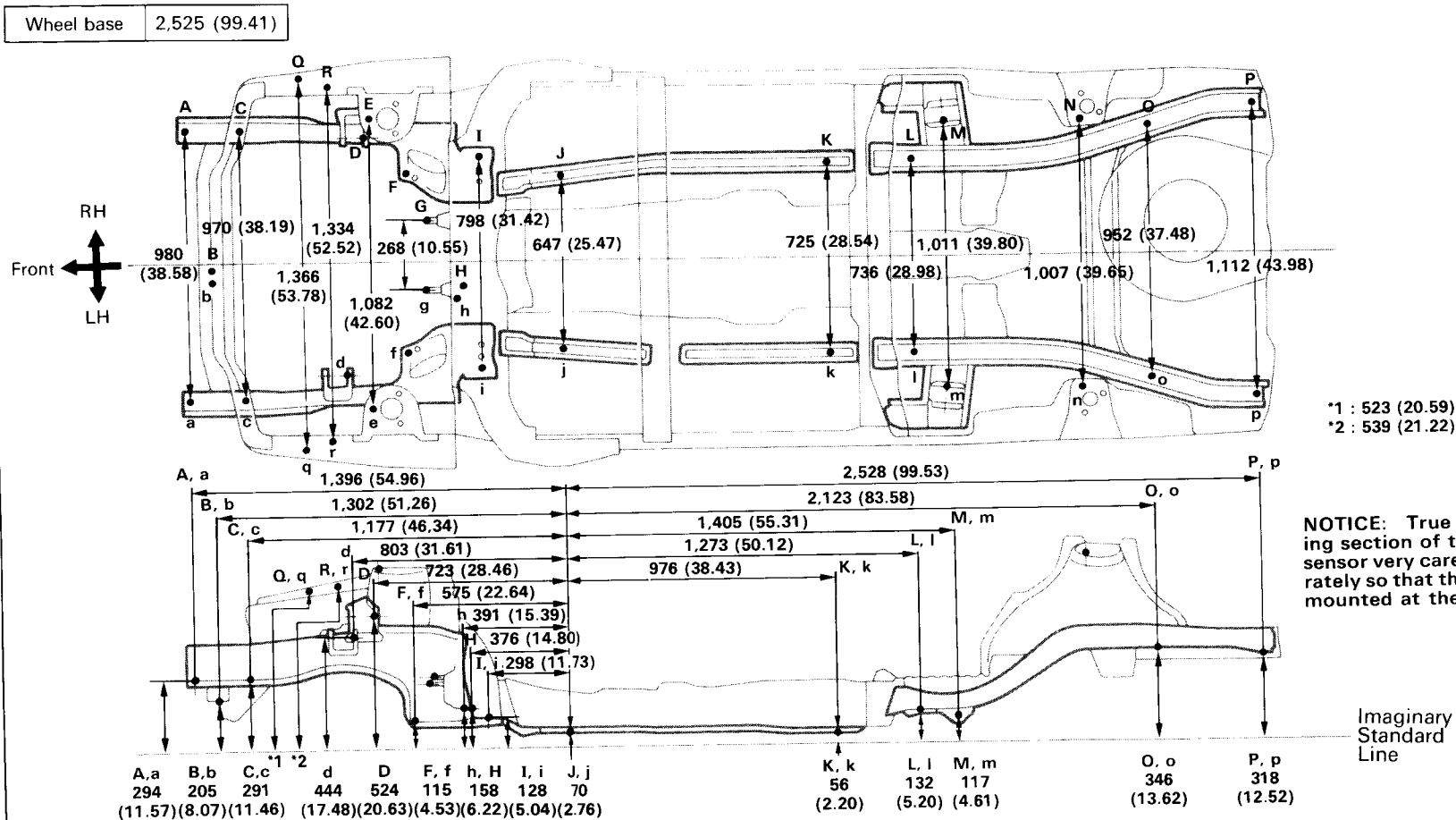


BO4465

Symbol	Name	Hole dia.	Symbol	Name	Hole dia.
A,a	Front side member bumper installation hole	RH 14 (0.55) LH 18 (0.71)	I,i	Lower arm installation nut	12 (0.47) nut
B,b	Engine mounting member installation nut	10 (0.39) nut	J,j	Front floor under reinforcement standard hole	15 (0.59)
C,c	Front side member standard hole	15 (0.59)	K,k	Front floor under reinforcement standard hole	15 (0.59)
D,d	Engine mounting bracket hole – rear	13 (0.51)	L,l	Rear floor side member standard hole	18 (0.71)
E,e	Front spring support hole – front	11 (0.43)	M,m	Strut bar installation hole – inner	12 (0.47)
F,f	Lower arm installation nut	14 (0.55) nut	N,n	Rear spring support hole – front = inner	9.5 (0.374)
G,g	Steering gear box installation bolt	12 (0.47) bolt	O,o	Rear floor side member standard hole	10 (0.39)
H,h	Engine mounting member installation nut	10 (0.39) nut	P,p	Rear floor side member standard hole	18 (0.71)
			Q,q-R,r	Front airbag sensor installation nut	8 (0.31)

UNDER BODY

(Two-Dimensional Distance)

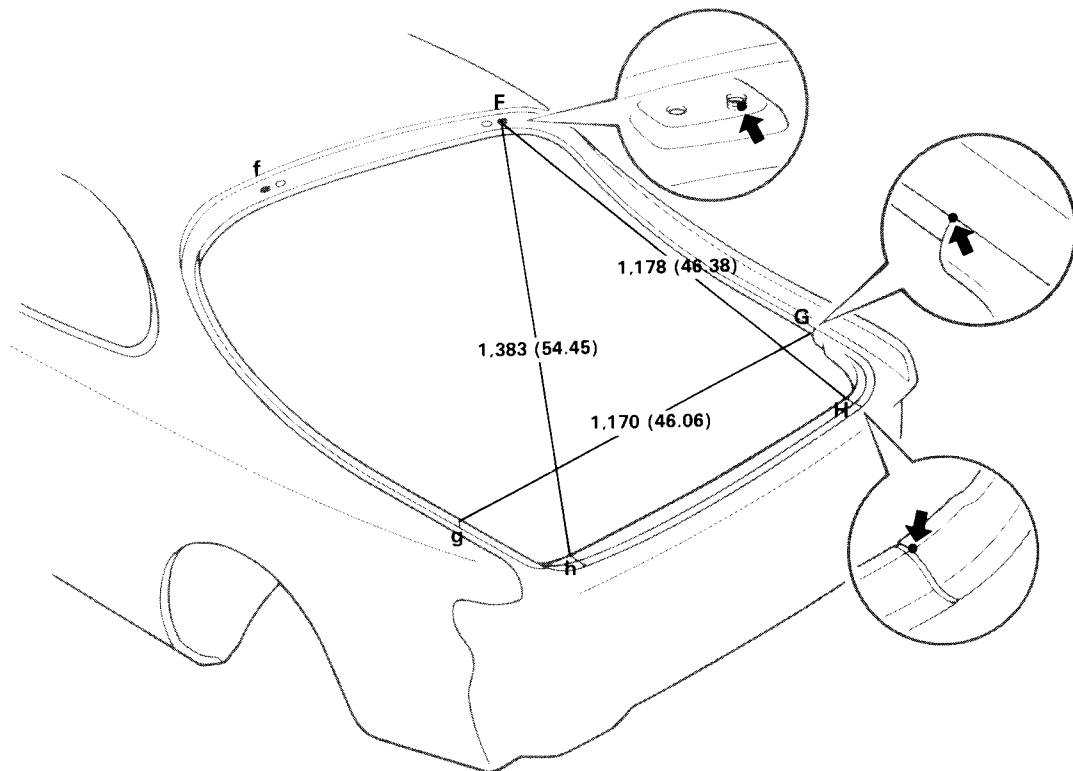


mm (in.)

Symbol	Name	Hole dia.	Symbol	Name	Hole dia.
A, a	Front side member bumper installation hole	RH 14 (0.55) LH 18 (0.71)	I, i	Lower arm installation nut	12 (0.47) nut
B, b	Engine mounting member installation nut	10 (0.39) nut	J, j	Front floor under reinforcement standard hole	15 (0.59)
C, c	Front side member standard hole	15 (0.59)	K, k	Front floor under reinforcement standard hole	15 (0.59)
D, d	Engine mounting bracket hole – rear	13 (0.51)	L, l	Rear floor side member standard hole	18 (0.71)
E, e	Front spring support hole – front	11 (0.43)	M, m	Strut bar installation hole – inner	12 (0.47)
F, f	Lower arm installation nut	14 (0.55) nut	N, n	Rear spring support hole – front = inner	9.5 (0.374)
G, g	Steering gear box installation bolt	12 (0.47) bolt	O, o	Rear floor side member standard hole	10 (0.39)
H, h	Engine mounting member installation nut	10 (0.39) nut	P, p	Rear floor side member standard hole	18 (0.71)
			Q, q, R, r	Front airbag sensor installation nut	8 (0.31)

LUGGAGE COMPARTMENT (Liftback)

(Three - Dimensional Distance)



mm (in.)

Symbol	Name	Hole dia.
F,f	Back door hinge installation hole - outer : rear end	11 (0.43)
G,g	Opening reinforcement/Quarter panel adjoining portion	—
H,h	Opening rein forcement/Lower back panel adjoining portion	—

SERVICE SPECIFICATIONS

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MAINTENANCE

Engine

Drive belt tension				
Alternator	w/ A/C	New belt	165 ± 10 lb	
		Used belt	84 ± 10 lb	
	w/o A/C	New belt	150 ± 25 lb	
		Used belt	130 ± 25 lb	
PS pump		New belt	125 ± 25 lb	
		Used belt	80 ± 20 lb	
Engine coolant capacity (w/ Heater)			6.4 liters	6.8 US qts 5.6 Imp. qts
Engine oil capacity (Drain and refill)			3.9 liters	3.7 US qts 3.4 Imp. qts
w/ Oil filter change			3.6 liters	3.8 US qts 3.2 Imp. qts
w/o Oil filter change				
Spark plug		ND	PK20R8	
Type		NGK	BKR6EP8	
			0.8 mm	0.031 in.
Air gap			1 – 3 – 4 – 2	
Firing order		Intake	0.15 – 0.25 mm	0.006 – 0.010 in.
Valve clearance		Exhaust	0.20 – 0.30 mm	0.008 – 0.012 in.

Chassis

Front and rear brakes				
Pad thickness		Limit	1.0 mm	0.039 in.
Disc thickness	Limit	Front	23.0 mm	0.906 in.
		Rear	9.0 mm	0.354 in.
Disc runout	Limit	Front	0.07 mm	0.0028 in.
		Rear	0.15 mm	0.0059 in.
Parking brake				
Lining thickness		Limit	1.0 mm	0.039 in.
Drum inside diameter		Limit	171 mm	6.732 in.
Front axle and suspension				
Ball joint vertical play		Limit	0 mm	0 in.
Steering wheel play			30 mm	1.18 in.
Torque specifications				
Front seat mount bolts			375 kg-cm	27 ft-lb 37 N·m
Engine mounting center member-to-body mount bolts			530 kg-cm	38 ft-lb 52 N·m
Front suspension lower crossmember-to-body mount bolts			1,550 kg-cm	112 ft-lb 152 N·m
Rear suspension lower crossmember-to-body mount bolts			1,280 kg-cm	93 ft-lb 126 N·m

ENGINE MECHANICAL

Specifications

Idle speed				800 ± 50 rpm	
Intake manifold vacuum		at Idle speed		450 mmHg	17.7 in.Hg 60 kPa
Compression		at 250 rpm	STD	11.5 kg/cm ² (164 psi, 1,128 kPa) or more	
			Limit	9.0 kg/cm ² 128 psi 883 kPa	
		Difference of pressure between each cylinder		1.0 kg/cm ² (14 psi, 98 kPa) or less	
Timing belt tensioner		Protrusion		8.5 – 9.5 mm	0.335 – 0.374 in.
Cylinder head	Warpage	Cylinder block side	Limit	0.20 mm	0.0079 in.
		Intake manifold side	Limit	0.20 mm	0.0079 in.
		Exhaust manifold side	Limit	0.30 mm	0.0118 in.
	Valve seat	Refacing angle		30°, 45°, 75°	
		Contacting angle		45°	
		Contacting width		1.0 – 1.4 mm	0.039 – 0.055 in.
Valve guide bushing	Inside diameter			6.000 – 6.018 mm	0.2362 – 0.2369 in.
	Outside diameter (for repair part)	STD		11.030 – 11.041 mm	0.4343 – 0.4347 in.
		O/S 0.05		11.080 – 11.91 mm	0.4362 – 0.4367 in.
Valve	Valve overall length	STD	Intake	105.50 mm	4.1535 in.
			Exhaust	99.55 mm	3.9193 in.
		Limit	Intake	104.80 mm	4.1260 in.
			Exhaust	98.85 mm	3.8917 in.
	Valve face angle			44.5°	
	Stem diameter		Intake	5.960 – 5.975 mm	0.2346 – 0.2352 in.
			Exhaust	5.955 – 5.970 mm	0.2344 – 0.2350 in.
	Stem oil clearance	STD	Intake	0.025 – 0.058 mm	0.0010 – 0.0023 in.
			Exhaust	0.030 – 0.063 mm	0.0012 – 0.0025 in.
		Limit	Intake	0.08 mm	0.0031 in.
			Exhaust	0.10 mm	0.0039 in.
	Margin thickness	STD		0.8 – 1.2 mm	0.031 – 0.047 in.
		Limit		0.5 mm	0.020 in.
Valve spring	Squareness	Limit		2.0 mm	0.079 in.
	Free length			44.43 mm	1.7492 in.
	Installed tension at 34.4 mm (1.354 in.)			20.5 – 24.1 kg (45.2 – 53.1 lb, 201 – 236 N)	
Valve lifter	Lifter diameter			27.975 – 27.985 mm	1.1014 – 1.1018 in.
	Lifter bore diameter			28.000 – 28.021 mm	1.1024 – 1.1032 in.
	Oil clearance	STD		0.015 – 0.046 mm	0.0006 – 0.0018 in.
		Limit		0.07 mm	0.0028 in.
Manifold	Warpage	Limit		0.20 mm	0.0079 in.

Specifications (Cont'd)

Camshaft	Thrust clearance	STD	0.120 – 0.240 mm	0.0047 – 0.0094 in.
		Limit	0.30 mm	0.0118 in.
	Journal oil clearance	STD	0.025 – 0.062 mm	0.0010 – 0.0024 in.
		Limit	0.08 mm	0.0031 in.
	Journal diameter		26.959 – 26.975 mm	1.0614 – 1.0620 in.
	Circle runout	Limit	0.06 mm	0.0024 in.
T-VIS valve	Cam lobe height	STD	41.010 – 41.110 mm	1.6146 – 1.6185 in.
		Limit	39.90 mm	1.5709 in.
	Warpage	Limit	0.20 mm	0.0079 in.
	Cylinder head surface warpage	Limit	0.05 mm	0.0020 in.
	Cylinder bore diameter	Mark 1	86.000 – 86.010 mm	3.3858 – 3.3862 in.
		Mark 2	86.010 – 86.020 mm	3.3862 – 3.3866 in.
Cylinder block		Mark 3	86.020 – 86.030 mm	3.3866 – 3.3870 in.
		Limit	86.23 mm	3.3949 in.
	Piston diameter	Mark 1	85.920 – 85.930 mm	3.3927 – 3.3831 in.
		Mark 2	85.930 – 85.940 mm	3.3831 – 3.3835 in.
		Mark 3	85.940 – 85.950 mm	3.3835 – 3.3839 in.
	Piston oil clearance	STD	0.070 – 0.090 mm	0.0028 – 0.0035 in.
Piston and piston ring		Limit	0.110 mm	0.0043 in.
	Piston ring groove clearance	No.1	0.040 – 0.080 mm	0.0016 – 0.0031 in.
		No.2	0.030 – 0.070 mm	0.0012 – 0.0028 in.
	Piston ring end gap	STD	0.330 – 0.550 mm	0.0130 – 0.0217 in.
		No.1	0.450 – 0.670 mm	0.0177 – 0.0264 in.
		No.2	0.200 – 0.600 mm	0.0079 – 0.0236 in.
Connecting rod		Oil	0.85 mm	0.0335 in.
		Limit	0.97 mm	0.0382 in.
		No.1	0.90 mm	0.0354 in.
		No.2		
		Oil		
Connecting rod	Thrust clearance	STD	0.160 – 0.312 mm	0.0063 – 0.0123 in.
		Limit	0.35 mm	0.35 in.
	Connecting rod bearing center wall thickness	STD	1.484 – 1.488 mm	0.0584 – 0.0586 in.
		Mark 1	1.488 – 1.492 mm	0.0586 – 0.0587 in.
		Mark 2	1.492 – 1.496 mm	0.0587 – 0.0589 in.
		Mark 3		
Connecting rod	Connecting rod oil clearance	STD	0.024 – 0.055 mm	0.0009 – 0.0022 in.
		STD	0.023 – 0.069 mm	0.0009 – 0.0027 in.
		U/S 0.25	0.08 mm	0.0031 in.
		Limit	0.05 mm	0.0020 in.
	Rod bending	Limit per 100 mm (3.94 in.)	0.15 mm	0.0059 in.
	Rod twist	Limit per 100 mm (3.94 in.)	22.005 – 22.017 mm	0.8663 – 0.8668 in.
Connecting rod	Bushing inside diameter		21.997 – 22.009 mm	0.8660 – 0.8665 in.
	Piston pin diameter		0.005 – 0.011 mm	0.0002 – 0.0004 in.
	Piston pin oil clearance	STD	0.05 mm	0.0020 in.
		Limit		

Specifications (Cont'd)

Crankshaft	Thrust clearance		STD	0.020 – 0.220 mm	0.0008 – 0.0087 in.
			Limit	0.30 mm	0.0118 in.
	Thrust washer thickness		STD	2.440 – 2.490 mm	0.0961 – 0.0980 in.
	Main journal oil clearance				
	STD	No.3	STD	0.025 – 0.044 mm	0.0010 – 0.0017 in.
			U/S 0.25	0.021 – 0.061 mm	0.0008 – 0.0024 in.
		Others		0.015 – 0.034 mm	0.0006 – 0.0013 in.
			U/S 0.25	0.029 – 0.069 mm	0.0011 – 0.0027 in.
			Limit	0.08 mm	0.0031 in.
	Main journal diameter		STD	54.988 – 55.003 mm	2.1653 – 2.1655 in.
			U/S 0.25	54.745 – 54.755 mm	2.1553 – 2.1557 in.
	Main bearing center wall thickness				
	STD	No.3	Mark 1	1.992 – 1.995 mm	0.0784 – 0.0785 in.
			Mark 2	1.995 – 1.998 mm	0.0785 – 0.0787 in.
			Mark 3	1.998 – 2.001 mm	0.0787 – 0.0788 in.
			Mark 4	2.001 – 2.004 mm	0.0788 – 0.0789 in.
			Mark 5	2.004 – 2.007 mm	0.0789 – 0.0790 in.
	STD	Others	Mark 1	1.997 – 2.000 mm	0.0786 – 0.0787 in.
			Mark 2	2.000 – 2.003 mm	0.0787 – 0.0789 in.
			Mark 3	2.003 – 2.006 mm	0.0789 – 0.0790 in.
			Mark 4	2.006 – 2.009 mm	0.0790 – 0.0891 in.
			Mark 5	2.009 – 2.012 mm	0.0791 – 0.0792 in.
	Crank pin diameter		STD	47.985 – 48.000 mm	1.8892 – 1.8898 in.
			U/S 0.25	47.745 – 47.755 mm	1.8797 – 1.8801 in.
	Circle runout		Limit	0.06 mm	0.0024 in.
	Main journal taper and out-of-round		Limit	0.02 mm	0.0008 in.
	Crank pin and out-of-round		Limit	0.02 mm	0.0008 in.

Torque Specifications

Part tightened		kg-cm	ft-lb	N-m
Oil pump pulley x Oil pump drive shaft		355	26	35
No.2 idler pulley x Cylinder block		440	32	43
No.1 idler pulley bracket x Cylinder head		440	32	43
Crankshaft pulley x Crankshaft		1,100	80	108
Camshaft timing pulley x Camshaft		600	43	59
		420	30	41
Timing belt tensioner x Cylinder head		210	15	21
Cylinder head x Cylinder block		500	36	49
		Turn 90°		
Camshaft bearing cap x Cylinder head		190	14	19
No.3 timing belt cover x Cylinder head		25	21 in.-lb	2.5
Cylinder head cover x Cylinder head		180	13	18
RH rear engine hanger x Cylinder head		195	14	19
Intake manifold x Cylinder head		195	14	19
Intake manifold stay x Intake manifold		260	19	25
Intake manifold stay x Cylinder block		260	19	25
Water by-pass pipe x Water pump cover		80	69 in.-lb	7.8

Torque Specifications (Cont'd)

Part tightened		kg-cm	ft-lb	N-m
		400	29	39
Water outlet x Cylinder head		195	14	19
EGR valve x Intake manifold		195	14	19
EGR pipe x Cylinder head		130	9	13
LH engine hanger x Cylinder head	12 mm head bolt	195	14	19
	14 mm head bolt	530	38	52
Exhaust manifold x Cylinder head		300	22	29
Catalytic converter x Turbine outlet elbow		600	43	59
Catalytic converter stay x Catalytic converter		400	29	39
No.1 alternator bracket x Cylinder head		600	43	59
Main bearing cap x Cylinder block		680	49	67
Connecting rod cap x Connecting rod		95	82 in.-lb	9.3
Rear oil seal retainer x Cylinder block		450	33	44
Knock sensor x Cylinder head		530	38	52
RH engine mounting bracket x Cylinder block		440	32	43
PS pump bracket x Cylinder block		95	82 in.-lb	9.3
Rear end plate x Cylinder block		1,100	80	108
Flywheel x Crankshaft		530	38	52
LH engine mounting bracket x Transaxle		650	47	63
LH engine mounting insulator x LH engine mounting bracket		890	64	87
LH engine mounting insulator x Body		530	38	52
RH engine mounting insulator x RH engine mounting bracket		890	64	87
RH engine mounting insulator x Body		740	54	73
RH engine mounting stay x RH engine mounting insulator		740	54	73
RH engine mounting stay x No.1 alternator bracket		210	15	21
LH engine mounting stay x LH engine mounting insulator		210	15	21
LH engine mounting stay x Transaxle		530	38	52
Front engine mounting bracket x Transaxle		790	57	77
Rear engine mounting bracket x Transaxle		530	38	52
Engine mounting center member x Body		740	54	73
Engine mounting center member x Front engine mounting insulator		740	54	73
Engine mounting center member x Rear engine mounting insulator		890	64	87
Front engine mounting insulator x Front engine mounting insulator		890	64	87
Rear engine mounting insulator x Rear engine mounting insulator		1,550	112	152
Suspension lower crossmember x Body		210	15	21
Suspension upper brace x Body	Bolt	650	47	64
	Nut			

EXHAUST SYSTEM**Torque Specifications**

Part tightened	kg-cm	ft-lb	N·m
Front exhaust pipe x Catalytic converter	630	46	62
Front exhaust pipe x Center exhaust pipe	440	32	43
Center exhaust pipe x Tailpipe	440	32	43

TURBOCHARGER SYSTEM**Specifications**

Turbocharger	Turbocharging pressure	0.50 – 0.83 kg/cm ² (7.1 – 11.8 psi, 49 – 81 kPa)
	Impeller wheel axial play	0.13 mm (0.0051 in.) or less
	Impeller wheel radial play	0.18 mm (0.0071 in.) or less

Torque Specifications

Part tightened		kg-cm	ft-lb	N·m
Turbine outlet elbow x Turbocharger		650	47	64
Side bearing housing plate x Turbocharger		120	9	11
Turbo water pipe x Turbocharger		120	9	11
Turbocharger x Exhaust manifold		650	47	64
Oil pipe x Turbocharger		175	13	17
Oil pipe x Cylinder block	Bolt	440	32	43
	Union bolt	525	38	51
Turbocharger stay x Turbocharger		705	51	69
Turbocharger stay x Cylinder block		600	43	59
Oxygen sensor x Turbine outlet elbow		450	33	44

EFI SYSTEM

Specifications

Fuel pressure regulator	Fuel pressure at No vacuum	2.3 – 2.7 kg/cm ² (33 – 38 psi, 226 – 265 kPa)	
Cold start injector	Resistance Fuel leakage	2 – 4 Ω One drop or less per minute	
Injector	Resistance Injection volume Difference between each cylinder Fuel leakage	2 – 4 Ω 95 – 120 cc (5.8 – 7.3 cu in.) per 15 sec. 5 cc (0.3 cu in.) or less One drop or less per minute	
Throttle body	Throttle body fully closed angle	6°	
Throttle position sensor	Clearance between stop screw and lever	Between terminals	Resistance
	0 mm 0 in. 0.50 mm 0.020 in. 0.70 mm 0.028 in. Throttle valve fully opened position –	VTA – E2 IDL – E2 IDL – E2 VTA – E2 VC – E2	0.2 – 0.8 kΩ Less than 2.3 kΩ Infinity 3.3 – 10.3 kΩ 3 – 8.3 kΩ
ISC valve	Resistance + B – RSC or RSO	19.3 – 22.3 Ω	
Cold start injector time switch	Resistance STA – STJ below 10°C (50°F) above 25°C (77°F) STA – Ground	30 – 50 Ω 70 – 90 Ω 30 – 90 Ω	
Air flow meter	Resistance VS – E2 VC – E2 THA – E2 at –20°C (–4°F) at 0°C (32°F) at 20°C (68°F) at 40°C (104°F) at 60°C (140°F)	200 – 600 Ω (Measuring plate fully closed) 20 – 1,200 Ω (Measuring plate fully open) 200 – 400 Ω 10 – 20 kΩ 4 – 7 kΩ 2 – 3 kΩ 0.9 – 1.3 kΩ 0.4 – 0.7 kΩ	
Fuel pump resistor	Resistance	Approx. 0.73 Ω	
Solenoid resistor	Resistance + B – No.10, No.20, No.30 or No.40	5 – 7 Ω	
Water temp.	Resistance at –20°C (–4°F) at 0°C (32°F) at 20°C (68°F) at 40°C (104°F) at 60°C (140°F) at 80°C (140°F)	10 – 20 kΩ 4 – 7 kΩ 2 – 7 kΩ 0.9 – 1.3 kΩ 0.4 – 0.7 kΩ 0.2 – 0.4 kΩ	
Oxygen sensor	Heater coil resistance	5.1 – 6.3 Ω	

Specifications (Cont'd)

T-VIS VSV	Resistance		33 – 39 Ω	
EGR VSV	Resistance		33 – 39 Ω	
Turbocharging pressure VSV	Resistance		24 – 30 Ω	
ECU	HINT: ● Perform all voltage and resistance measurements with the ECU connected. ● Verify that the battery voltage is 11 V or above with the ignition switch is ON.			
	Voltage			
	Terminals	Condition		STD voltage (V)
	+B – E1 +B1	IG SW ON		10 – 14
	BATT – E1	–		10 – 14
	IDL – E2	IG SW ON	Throttle valve open	4.5 – 5.5
	VTA – E2		Throttle valve fully closed	0.1 – 1.0
			Throttle valve open	3.2 – 4.2
	VC – E2		–	4.5 – 5.5
	VS – E2		Measuring plate fully closed	3.7 – 4.3
			Measuring plate fully open	0.2 – 0.5
		Idling	1.6 – 4.1	
		3,000 rpm	1.0 – 2.0	
	No.1 No.2 – E01 No.3 – E02 No.4	IG SW ON		10 – 14
	THA1 – E2	IG SW ON	Intake air temp. 20°C (68°F)	1 – 3
	THW – E2		Coolant temp. 80°C (176°F)	0.1 – 1.1
	STA – E1	Cranking		6 – 14
	IGT – E1	Cranking or idling		0.8 – 1.2
	RSC – E1 RSO	IG SW ON		8 – 14
	W – E1	No trouble ("CHECK" engine warning light off) and engine running		10 – 14
	PIM – E2	IG SW ON		2.5 – 4.5
	AC1 – E1	IG SW ON	Air conditioning ON	8 – 14
	ACT – E1		Air conditioning ON	4.5 – 5.5

Specifications (Cont'd)

Specifications (Cont'd)

ECU (cont'd)	Voltage (cont'd)			STD voltage (V)
	Terminals	Condition		
*1 TVIS – E1	IG SW ON	Throttle valve fully closed		2.0 or less
		Throttle valve open		10 – 14
*2 TVIS – E1	Idling			2.0 or less
	4,200 rpm or more			10 – 14
T – E1	IG SW ON	Check connector TE1 – E1 not connected		10 – 14
		Check connector TE1 – E1 connected		0.5 or less
Resistance				
	Terminals	Condition		STD resistance (Ω)
IDL – E2		Throttle valve open		Infinity
		Throttle valve fully closed		2,300 Ω or less
VTA – E2		Throttle valve fully open		3,500 – 10,000
		Throttle valve fully closed		200 – 800
VC – E2		–		200 – 400
VS – E2		Measuring plate fully closed		200 – 600
		Measuring plate fully open		20 – 1,200
THA – E2		Intake air temp. 20°C (68°F)		2,000 – 3,000
THW – E2		Coolant temp. 80°C (176°F)		200 – 400
G1 – G ⊖ G2		–		140 – 180
NE – G ⊖		–		180 – 220
RSC + B RSO + B1		–		19.3 – 22.3
Fuel cut rpm	w/ Vehicle speed 0 km/h and coolant and coolant temp. 80°C (176°F)			
	Fuel cut rpm			2,000 rpm
	Fuel return rpm			1,600 rpm

*¹ w/ Regular Unleaded Gasoline *² w/ Premium Unleaded Gasoline

Torque Specifications

Part tightened		kg-cm	ft-lb	N-m
Fuel line	Union bolt type	300	22	29
	Flare nut type	310	22	30
		30	26 in.-lb	2.9
Fuel pump x Fuel tank		15	13 in.-lb	1.5
Fuel sender gauge x Fuel tank				

Torque Specifications (Cont'd)

Part tightened	kg-cm	ft-lb	N·m
Fuel evaporation bent tube x Fuel tank	15	13 in.-lb	1.5
Fuel tank band x Body	220	16	2.9
Fuel inlet pipe x Fuel tank	30	26 in.-lb	2.9
Cold start injector x Air intake chamber	60	52 in.-lb	5.9
Cold start injector pipe x Cold start injector	180	13	18
Cold start injector pipe x Delivery pipe	180	13	18
Fuel pressure regulator x Delivery pipe	300	22	29
Delivery pipe x Cylinder head	195	14	19
Injector cover x Delivery pipe	80	69 in.-lb	7.8
Fuel inlet hose x Delivery pipe	Bolt Union bolt	69 in.-lb	7.8
		22	29
Fuel inlet hose x Fuel filter	300	22	29
Throttle body x Intake manifold	195	14	19
Intake air connector stay x Throttle body	195	14	19
Intake air connector stay x Cylinder head	80	69 in.-lb	7.8
Intake air connector x Throttle body	195	14	19

COOLING SYSTEM

Specifications

Engine coolant capacity (w/ Heater)	6.4 liters	6.8 US qts	5.6 Imp. qts
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Torque Specifications

Part tightened	kg-cm	ft-lb	N·m
Engine coolant x Drain plug	130	9	13
Water pump x Water pump cover	95	82 in.-lb	9.3

LUBRICATION SYSTEM

Specifications

Engine oil capacity		See page A-25
Oil pressure	at Idling at 3,000 rpm	0.3 kg/cm ² (4.3 psi, 29 kPa) or more 2.5 – 5.0 kg/cm ² (36 – 71 psi, 245 – 490 kPa)

Torque Specifications

Part tightened	kg-cm	ft-lb	N·m
Engine oil drain plug	250	18	25
Oil cooler bracket x Cylinder head	80	69 in.-lb	7.8
Oil cooler x Cylinder block	800	58	78
Water by-pass pipe x Oil cooler	120	9	12
Water by-pass pipe x Oil cooler bracket	180	13	18
Oil nozzle x Cylinder block	93	81 in.-lb	9.1

IGNITION SYSTEM

Ignition timing		10° BTDC @ idle (w/ Terminals TE1 and E1 connected)
Firing order		1 – 3 – 4 – 2
Spark plug		See page A-2
High-tension cord	Resistance	25 k Ω per cord
Ignition coil	Primary coil resistance	0.41 – 0.50 Ω
	Secondary coil resistance	10.2 – 13.8 k Ω
Distributor	Air gap	0.2 – 0.4 mm 0.008 – 0.016 in.
	Signal generator (pickup coil) resistance	
	G1 – G ⊖	140 – 180 Ω
	G2 – G ⊖	140 – 180 Ω
	NE – G ⊖	180 – 220 Ω

CHARGING SYSTEM

Drive belt tension	See page A-2
Battery specific gravity when full charge at 20°C (68°F)	1.25 – 1.27

CLUTCH**Specifications**

Disc rivet head depth	Limit	0.3 mm	0.012 in.
Disc runout	Limit	0.8 mm	0.031 in.
Diaphragm spring out of alignment	Limit	0.5 mm	0.020 in.
Diaphragm spring finger wear	Depth	0.6 mm	0.024 in.
	Width	5.0 mm	0.197 in.
Flywheel runout	Limit	0.1 mm	0.004 in.

Torque Specifications

Part tightened	kg-cm	ft-lb	N-m
Clutch cover x Flywheel	195	14	19
Release fork support	480	35	47

MANUAL TRANSAXLE (E150F)**Specifications**

TRANSMISSION ASSEMBLY				
Input shaft				
Roller bearing journal diameter		Limit	32.930 mm	1.2964 in.
3rd gear journal diameter		Limit	35.950 mm	1.4154 in.
4th gear journal diameter		Limit	35.950 mm	1.4154 in.
Runout		Limit	0.060 mm	0.0024 in.
Output shaft				
1st gear journal diameter		Limit	38.950 mm	1.5335 in.
2nd gear journal diameter		Limit	38.950 mm	1.5335 in.
Runout		Limit	0.060 mm	0.0024 in.
Gear thrust clearance	1st	STD	0.10 – 0.35 mm	0.0039 – 0.0138 in.
		Limit	0.40 mm	0.0157 in.
	2nd	STD	0.10 – 0.45 mm	0.0039 – 0.0177 in.
		Limit	0.50 mm	0.0197 in.
	3rd	STD	0.10 – 0.35 mm	0.0039 – 0.0139 in.
		Limit	0.40 mm	0.0157 in.
	4th	STD	0.10 – 0.55 mm	0.0039 – 0.0217 in.
		Limit	0.60 mm	0.0236 in.
	5th	STD	0.10 – 0.57 mm	0.0039 – 0.0224 in.
		Limit	0.65 mm	0.0256 in.
Gear oil clearance				
1st and 4th gear	STD	0.009 – 0.051 mm	0.0004 – 0.0020 in.	
	Limit	0.080 mm	0.0031 in.	
2nd and 3rd gear	STD	0.009 – 0.053 mm	0.0004 – 0.0020 in.	
	Limit	0.080 mm	0.0031 in.	
5th gear	STD	0.009 – 0.050 mm	0.0004 – 0.0020 in.	
	Limit	0.070 mm	0.0028 in.	
Shift fork to hub sleeve clearance		Limit	1.0 mm	0.039 in.
Synchronizer ring to gear clearance		Limit	0.6 mm	0.024 in.
Oil pump assembly				
Rotor body clearance	STD	0.10 – 0.16 mm	0.0039 – 0.0063 in.	
	Limit	0.30 mm	0.0118 in.	
Rotor tip clearance	STD	0.08 – 0.15 mm	0.0031 – 0.0059 in.	
	Limit	0.30 mm	0.0118 in.	
Side clearance	STD	0.03 – 0.08 mm	0.0012 – 0.0031 in.	
	Limit	0.15 mm	0.0059 in.	
Oil seal drive in depth				
Control shaft cover			0 – 1.0 mm	0 – 0.039 in.
Input shaft snap ring thickness				
No.2 clutch hub	Mark			
	H		2.30 – 2.35 mm	0.0906 – 0.0925 in.
	J		2.35 – 2.40 mm	0.0925 – 0.0945 in.
	K		2.40 – 2.45 mm	0.0945 – 0.0965 in.
	L		2.45 – 2.50 mm	0.0965 – 0.0984 in.
	M		2.50 – 2.55 mm	0.0984 – 0.1004 in.
	N		2.55 – 2.60 mm	0.1004 – 0.1024 in.
P		2.60 – 2.65 mm	0.1024 – 0.1043 in.	

Specifications (Cont'd)

Input shaft snap ring thickness (cont'd)			
4th gear and rear bearing	Mark		
	1	2.35 – 2.40 mm	0.0925 – 0.0945 in.
	2	2.40 – 2.45 mm	0.0945 – 0.0965 in.
	3	2.45 – 2.50 mm	0.0965 – 0.0984 in.
	4	2.50 – 2.55 mm	0.0984 – 0.1004 in.
	5	2.55 – 2.60 mm	0.1004 – 0.1024 in.
	6	2.60 – 2.65 mm	0.1024 – 0.1043 in.
	7	2.65 – 2.70 mm	0.1043 – 0.1063 in.
	8	2.70 – 2.75 mm	0.1063 – 0.1083 in.
No.3 clutch hub		Mark	
	Q	2.25 – 2.30 mm	0.0886 – 0.0906 in.
	R	2.30 – 2.35 mm	0.0906 – 0.0925 in.
	S	2.35 – 2.40 mm	0.0925 – 0.0945 in.
	T	2.40 – 2.45 mm	0.0945 – 0.0965 in.
	U	2.45 – 2.50 mm	0.0965 – 0.0984 in.
	V	2.50 – 2.55 mm	0.0984 – 0.1004 in.
	W	2.55 – 2.60 mm	0.1004 – 0.1024 in.
	X	2.60 – 2.65 mm	0.1024 – 0.1043 in.
	Y	2.65 – 2.70 mm	0.1043 – 0.1063 in.
Output shaft snap ring thickness			
No.1 clutch hub	Mark		
	A	2.80 – 2.85 mm	0.1102 – 0.1122 in.
	B	2.85 – 2.90 mm	0.1122 – 0.1142 in.
	C	2.90 – 2.95 mm	0.1142 – 0.1161 in.
	D	2.95 – 3.00 mm	0.1161 – 0.1181 in.
	E	3.00 – 3.05 mm	0.1181 – 0.1201 in.
	F	3.05 – 3.10 mm	0.1201 – 0.1220 in.
	G	3.10 – 3.15 mm	0.1220 – 0.1240 in.
DIFFERENTIAL ASSEMBLY			
Output shaft preload (at starting)	New bearing	8.0 – 16.0 kg-cm	6.9 – 13.9 in.-lb 0.78 – 1.57 N·m
	Reused bearing	5.0 – 10.0 kg-cm	4.3 – 8.7 in.-lb 0.49 – 0.98 N·m
Differential case preload (at starting)	New bearing	1.9 – 3.7 kg-cm	1.6 – 3.2 in.-lb 0.2 – 0.4 N·m
	Reused bearing	1.2 – 2.3 kg-cm	1.0 – 2.0 in.-lb 0.1 – 0.2 N·m
Left case inner diameter			
Right case bushing	STD	111.000 – 111.035 mm	4.3701 – 4.3714 in.
	Limit	110.060 mm	4.3724 in.
Left case bushing	STD	90.500 – 90.535 mm	3.5630 – 3.5644 in.
	Limit	90.560 mm	3.5653 in.
No.2 differential case outer diameter			
Right side	STD	110.929 – 110.964 mm	4.3673 – 4.3687 in.
	Limit	110.850 mm	4.3632 in.
Left side	STD	90.429 – 90.464 mm	3.5602 – 3.5616 in.
	Limit	90.350 mm	3.5571 in.
Differential No.2 case inner diameter			
Side gear bushing	STD	35.000 – 35.025 mm	1.3780 – 1.3789 in.
	Limit	35.030 mm	1.3791 in.

Specifications (Cont'd)

Conical spring washer height			
Left conical spring washer	STD	2.60 – 2.80 mm	0.1024 – 0.1102 in.
	Limit	2.50 mm	0.0984 in.
Right conical spring washer	STD	1.70 – 1.90 mm	0.0669 – 0.0748 in.
	Limit	1.60 mm	0.0630 in.
Right case backlash		0.05 – 0.20 mm	0.0020 – 0.0079 in.
No.2 differential case backlash		0.05 – 0.20 mm	0.0020 – 0.0079 in.
Intermediate case backlash		0.05 – 0.20 mm	0.0020 – 0.0079 in.
Front differential side gear backlash		0.14 – 0.21 mm	0.0055 – 0.0083 in.
Thrust washer thickness			
For right case side gear		0.80 mm	0.0315 in.
		0.85 mm	0.0335 in.
		0.90 mm	0.0354 in.
		0.95 mm	0.0374 in.
		1.00 mm	0.0394 in.
		1.05 mm	0.0413 in.
		1.10 mm	0.0433 in.
		1.15 mm	0.0453 in.
		1.20 mm	0.0472 in.
		1.25 mm	0.0492 in.
		1.30 mm	0.0512 in.
		1.35 mm	0.0531 in.
		1.40 mm	0.0551 in.
For intermediate case side gear		0.80 mm	0.0315 in.
		0.85 mm	0.0335 in.
		0.90 mm	0.0354 in.
		0.95 mm	0.0374 in.
		1.00 mm	0.0394 in.
		1.05 mm	0.0413 in.
		1.10 mm	0.0433 in.
		1.15 mm	0.0453 in.
		1.20 mm	0.0472 in.
		1.25 mm	0.0492 in.
		1.30 mm	0.0512 in.
		1.35 mm	0.0531 in.
For No.2 differential case side gear		1.40 mm	0.0551 in.
	Mark		
	B	1.00 mm	0.0394 in.
	C	1.05 mm	0.0413 in.
	D	1.10 mm	0.0433 in.
	E	1.15 mm	0.0453 in.
	F	1.20 mm	0.0472 in.
For left case side gear		1.25 mm	0.0492 in.
	Mark		
	A	0.95 mm	0.0374 in.
	B	1.00 mm	0.0394 in.
	C	1.05 mm	0.0413 in.
	D	1.10 mm	0.0433 in.
	E	1.15 mm	0.0453 in.

Specifications (Cont'd)

Thrust washer thickness (cont'd)			
For left case side gear (cont'd)	Mark		
	F	1.20 mm	0.0472 in.
	G	1.25 mm	0.0492 in.
	H	1.30 mm	0.0512 in.
	J	1.35 mm	0.0531 in.
	K	1.40 mm	0.0551 in.
Adjusting shim thickness			
For differential preload	Mark		
	0	2.00 mm	0.0787 in.
	1	2.05 mm	0.0807 in.
	2	2.10 mm	0.0827 in.
	3	2.15 mm	0.0846 in.
	4	2.20 mm	0.0866 in.
	5	2.25 mm	0.0886 in.
	6	2.30 mm	0.0906 in.
	7	2.35 mm	0.0925 in.
	8	2.40 mm	0.0945 in.
	9	2.45 mm	0.0965 in.
	A	2.50 mm	0.0984 in.
	B	2.55 mm	0.1004 in.
	C	2.60 mm	0.1024 in.
	D	2.65 mm	0.1043 in.
	E	2.70 mm	0.1063 in.
	F	2.75 mm	0.1083 in.
	G	2.80 mm	0.1102 in.
	H	2.85 mm	0.1122 in.
For output shaft preload	Mark		
	0	0.40 mm	0.0157 in.
	1	0.45 mm	0.0177 in.
	2	0.50 mm	0.0197 in.
	3	0.55 mm	0.0217 in.
	4	0.60 mm	0.0236 in.
	5	0.65 mm	0.0256 in.
	6	0.70 mm	0.0276 in.
	7	0.75 mm	0.0295 in.
	8	0.80 mm	0.0315 in.
	9	0.85 mm	0.0335 in.
	A	0.90 mm	0.0354 in.
	B	0.95 mm	0.0374 in.
	C	1.00 mm	0.0394 in.
	D	1.05 mm	0.0413 in.
	E	1.10 mm	0.0433 in.
	F	1.15 mm	0.0453 in.
	G	1.20 mm	0.0472 in.
	H	1.25 mm	0.0492 in.
	J	1.30 mm	0.0512 in.
	K	1.35 mm	0.0531 in.
	L	1.40 mm	0.0551 in.

Specifications (Cont'd)

Adjusting shim thickness (cont'd)				
For output shaft preload (cont'd)	M	1.45 mm		0.0571 in.
	N	1.50 mm		0.0591 in.
	P	1.55 mm		0.0610 in.
	Q	1.60 mm		0.0630 in.
TRANSFER ASSEMBLY				
Ring gear backlash		0.13 – 0.18 mm		0.0051 – 0.0071 in.
Ring gear runout	Limit	0.1 mm		0.004 in.
Driven pinion preload (at starting)	New bearing	1.8 – 2.9 kg	4.0 – 6.4 lb	17.7 – 28.4 N
	Reused bearing	0.9 – 1.4 kg	2.0 – 3.1 lb	8.8 – 13.7 N
Transfer total preload (at starting)				
(Add driven pinion preload)	New bearing	1.3 – 1.4 kg	2.9 – 3.1 lb	12.7 – 13.7 N
	Reused bearing	0.5 – 0.9 kg	1.1 – 2.0 lb	4.9 – 8.8 N
Right case bushing inner diameter	STD	69.000 – 69.035 mm		2.7165 – 2.7179 in.
	Limit	69.060 mm		2.7189 in.
Left case bushing inner diameter	STD	69.000 – 69.035 mm		2.7165 – 2.7179 in.
	Limit	69.060 mm		2.7189 in.
Control coupling washer thickness	STD	1.49 – 1.51 mm		0.0587 – 0.0594 in.
	Limit	1.45 mm		0.0571 in.
Oil seal drive in depth				
Extension housing		1.1 – 1.9 mm		0.043 – 0.075 in.
Differential lock shift lever shaft		1.0 – 2.0 mm		0.039 – 0.079 in.
Shift fork shaft		8.5 – 9.5 mm		0.335 – 0.374 in.
Adjusting shim thickness				
For ring gear backlash	Mark			
	1	2.13 mm		0.0839 in.
	2	2.16 mm		0.0850 in.
	3	2.19 mm		0.0862 in.
	4	2.22 mm		0.0874 in.
	5	2.25 mm		0.0886 in.
	6	2.28 mm		0.0898 in.
	7	2.31 mm		0.0909 in.
	8	2.34 mm		0.0921 in.
	9	2.37 mm		0.0933 in.
	10	2.40 mm		0.0945 in.
	11	2.43 mm		0.0957 in.
	12	2.46 mm		0.0968 in.
	13	2.49 mm		0.0980 in.
	14	2.52 mm		0.0992 in.
	15	2.55 mm		0.1004 in.
	16	2.58 mm		0.1016 in.
	17	2.61 mm		0.1028 in.
	18	2.64 mm		0.1039 in.
	19	2.67 mm		0.1051 in.
	20	2.70 mm		0.1063 in.
	21	2.73 mm		0.1075 in.
	22	2.76 mm		0.1087 in.
	23	2.79 mm		0.1098 in.
	24	2.82 mm		0.1110 in.

Specifications (Cont'd)

Adjusting shim thickness (cont'd)			
For tooth contact	Mark		
	A	0.30 mm	0.0118 in.
	B	0.33 mm	0.0130 in.
	C	0.36 mm	0.0142 in.
	D	0.39 mm	0.0154 in.
	E	0.42 mm	0.0165 in.
	F	0.45 mm	0.0177 in.
	G	0.48 mm	0.0189 in.
	H	0.51 mm	0.0201 in.
	J	0.54 mm	0.0213 in.
	K	0.57 mm	0.0224 in.
Shift lever preload		60 – 150 g	0.1 – 0.3 lb 0.49 – 1.47 N
Shift lever seat shim thickness	Mark		
	A	0.5 mm	0.020 in.
	B	0.6 mm	0.024 in.
	C	0.7 mm	0.028 in.
	D	0.8 mm	0.031 in.
	E	0.9 mm	0.035 in.
	F	1.0 mm	0.039 in.
	G	1.1 mm	0.043 in.
	H	1.2 mm	0.047 in.
	J	1.3 mm	0.051 in.
	K	1.4 mm	0.055 in.
	L	1.5 mm	0.059 in.
	M	1.6 mm	0.063 in.
	N	1.7 mm	0.067 in.

Torque Specifications

Part tightened	kg-cm	ft-lb	N-m
Oil pump x Cover	105	8	10
Oil cooler pipe x Union	350	25	34
Ring gear x Differential case	1,260	91	124
Union x Transaxle	275	20	27
Rear bearing retainer x Transmission case	430	31	42
Straight screw plug	250	18	25
Transmission case x Transaxle case	300	22	29
Reverse restrict pin holder plug	130	9	13
Oil pump assembly x Transaxle case	175	13	17
Control shaft lock bolt	500	36	49
Shift fork x Fork shaft	240	17	24
Reverse idler gear shaft lock bolt	300	22	29
Oil pipe x Housing	175	13	17
Rear bearing retainer x Transmission case	430	31	42
Transmission case x Transmission case cover	300	22	29
5th driven gear lock nut	1,250	90	123
Shift and select lever assembly x Transmission case	200	14	20

Torque Specifications (Cont'd)

Part tightened		kg-cm	ft-lb	N·m
Transaxle case receiver x Transaxle case		75	65 in.-lb	7.4
Bellcrank x Transmission case		200	14	20
Shift arm bracket bolt x Transaxle case		175	13	17
Back-up light switch x Transmission case		410	30	40
Differential right case x Intermediate case		640	46	63
Driven pinion x Bearing cage		See page MT-78		
Ring gear x Ring gear mounting		985	71	97
Driven pinion cage x Transfer case		400	29	39
Transfer case x Transfer right case		450	33	44
Transfer case x Inspection hole cover		160	12	16
Transfer shift lever shaft x Left case		200	14	20
Differential lock sleeve x Shift fork		160	12	16
Transfer right case x Transfer case cover		175	13	17
Transfer oil tube set bolt		130	9	13
Driven pinion cage x Extension housing		260	19	25
Differential lock shift fork shaft lock screw		400	29	39
Extension housing x Dynamic damper		260	19	25
Transaxle assembly x Transfer assembly		700	51	69
Drain plug		400	29	39
Transaxle assembly x Engine	M12 bolt	650	47	64
	M10 bolt	470	34	46
LH engine mounting bracket x Transaxle		530	38	52
Mounting insulator x Mounting bracket		890	64	87
Mounting bracket x Transaxle bracket		650	47	63
RH mounting insulator x Bracket		530	38	52
RH mounting stay bolt		740	54	73
LH mounting stay bolt		210	15	21
Catalytic converter mounting bolt		300	22	29
Converter stay set bolt		600	43	59
Front mounting bracket x Transaxle		790	57	77
Rear mounting bracket x Transaxle		790	57	77
Center member x Body		530	38	52
Center member x Mounting bracket		740	54	73
A/C compressor x Idler pulley bracket		280	20	27
Exhaust pipe x Converter		630	46	62
Exhaust pipe x Center exhaust pipe		440	32	43
Lower crossmember x Body		1,500	112	152
Fuel hose union bolt		300	22	29
Suspension upper brace x Body	Bolt	210	15	21
	Nut	650	47	64
Shift lever retainer x Plate		120	9	12
Selecting spring cover x Shift lever retainer		50	43 in.-lb	4.9
Shift lever retainer plate x Body		120	9	12

PROPELLER SHAFT**Specifications**

Bearing axial play		Limit	0.05 mm	0.0020 in.
Front propeller shaft runout		Limit	0.8 mm	0.031 in.
Intermediate shaft runout		Limit	0.8 mm	0.031 in.
Rear propeller shaft runout		Limit	0.8 mm	0.031 in.
Intermediate shaft flange runout		Limit	0.1 mm	0.0039 in.
Snap ring thickness	Color	None	1.475 – 1.525 mm	0.0581 – 0.0600 in.
		Brown	1.525 – 1.575 mm	0.0600 – 0.0620 in.
		Blue	1.575 – 1.625 mm	0.0620 – 0.0640 in.

Torque Specifications

Part tightened		kg-cm	ft-lb	N·m
Propeller shaft x Differential		750	54	74
Intermediate shaft x Propeller shaft		750	54	74
Center support bearing x Body		375	27	37
Intermediate shaft x Center support bearing x Joint flange		1,850	134	181
	1st	Loosen nut		
	2nd			
	3rd	700	51	69
Cross groove joint set bolt		660	48	65

SUSPENSION AND AXLE

Specifications

Cold tire inflation pressure kg/cm ² (psi, kPa)	Tire size	Front	Rear		
	215/50R15 88V	2.2 (32, 210)	2.1 (30, 206)		
Chassis ground clearance mm (in.)	All models	191.0 (7.520)	233.5 (9.193)		
Rear wheel alignment		Inspection STD	Adjustment STD		
	Toe-in	5 ± 2 mm (0.20 ± 0.08 in.)	5 ± 1 mm (0.20 ± 0.04 in.)		
	Left-right error	3.0 mm (0.12 in.) or less	3.0 mm (0.12 in.) or less		
	Camber	–45' ± 30'	–		
	Left-right error	30' or less	–		
Wheel lateral runout		Less than 1.0 mm (0.039 in.)			
Front axle and suspension	Hub bearing axial direction play	Limit	0.05 mm	0.0020 in.	
	Ball joint rotation condition		10 – 30 kg-cm	9 – 26 in.-lb	1.0 – 2.9 N·m
	Ball joint vertical play		0 mm	0 in.	
	Drive shaft grease capacity				
	Outboard joint		120 g	0.26 lb	
	Inboard joint		90 g	0.20 lb	
	Side gear shaft		43 g	0.09 lb	
	Drive shaft standard length		405.4 mm	15.96 in.	
Rear axle and suspension	Hub bearing axial direction play	Limit	0.05 mm	0.0020 in.	
	Rear axle hub runout	Limit	0.07 mm	0.0028 in.	
	Stabilizer bar link				
	Ball joint rotation condition		0.5 – 10.0 kg-cm	(0.434 – 8.680 in.-lb, 0.049 – 0.981 N·m)	
	Drive shaft grease capacity				
	Inboard joint		180 g	0.40 lb	
	Outboard joint		120 g	0.26 lb	
	Drive shaft standard length		558.5 mm	21.99 in.	
Differential	Drive pinion bearing preload (at Starting)				
	New bearing		10 – 16 kg-cm	(8.7 – 13.9 in.-lb, 1.0 – 1.6 N·m)	
	Reused bearing		5 – 8 kg-cm	(4.3 – 6.9 in.-lb, 0.5 – 0.8 N·m)	
	Total preload (at Starting)		Add drive pinion bearing preload		
			3 – 5 kg-cm (2.6 – 4.3 in.-lb, 0.3 – 0.5 N·m)		
	Drive pinion to ring gear backlash		0.13 – 0.18 mm	0.0051 – 0.0071 in.	
	Side gear backlash		0.05 – 0.20 mm	0.0020 – 0.0079 in.	
Ring gear runout	Limit	0.07 mm	0.0028 in.		

Specifications (Cont'd)

Differential (cont'd)	Companion flange runout		0.1 mm	0.004 in.
	Lateral runout	Limit	0.1 mm	0.004 in.
	Radial runout	Limit	90 – 110°C	194 – 230°F
	Ring gear installing temperature		2.0 mm	0.079 in.
	Drive pinion oil seal drive in depth		Flash the carrier end surface	
	Side gear oil seal drive in depth		0.95 mm	0.0374 in.
	Side gear thrust washer thickness		1.00 mm	0.0394 in.
			1.05 mm	0.0413 in.
			1.10 mm	0.0433 in.
			1.15 mm	0.0453 in.
			1.20 mm	0.0472 in.
	Side bearing adjusting washer thickness		2.21 – 2.23 mm	0.0870 – 0.0878 in.
			2.24 – 2.26 mm	0.0882 – 0.0890 in.
			2.27 – 2.29 mm	0.0894 – 0.0902 in.
			2.30 – 2.32 mm	0.0906 – 0.0913 in.
			2.33 – 2.35 mm	0.0917 – 0.0925 in.
			2.36 – 2.38 mm	0.0929 – 0.0937 in.
			2.39 – 2.41 mm	0.0941 – 0.0949 in.
			2.42 – 2.44 mm	0.0953 – 0.0961 in.
			2.45 – 2.47 mm	0.0965 – 0.0972 in.
			2.48 – 2.50 mm	0.0976 – 0.0984 in.
			2.51 – 2.53 mm	0.0988 – 0.0996 in.
			2.54 – 2.56 mm	0.1000 – 0.1008 in.
			2.57 – 2.59 mm	0.1012 – 0.1020 in.
			2.60 – 2.62 mm	0.1024 – 0.1031 in.
			2.63 – 2.65 mm	0.1035 – 0.1043 in.
			2.66 – 2.68 mm	0.1047 – 0.1055 in.
			2.69 – 2.71 mm	0.1059 – 0.1067 in.
			2.72 – 2.74 mm	0.1071 – 0.1079 in.
			2.75 – 2.77 mm	0.1083 – 0.1091 in.
			2.78 – 2.80 mm	0.1094 – 0.1102 in.
			2.81 – 2.83 mm	0.1106 – 0.1114 in.
			2.84 – 2.86 mm	0.1118 – 0.1126 in.
			2.87 – 2.89 mm	0.1130 – 0.1138 in.
			2.90 – 2.92 mm	0.1142 – 0.1150 in.
			2.93 – 2.95 mm	0.1154 – 0.1161 in.
			2.96 – 2.98 mm	0.1165 – 0.1173 in.
			2.99 – 3.01 mm	0.1177 – 0.1185 in.
			3.02 – 3.04 mm	0.1189 – 0.1197 in.
			3.05 – 3.07 mm	0.1201 – 0.1209 in.
			3.08 – 3.10 mm	0.1213 – 0.1220 in.
			3.11 – 3.13 mm	0.1224 – 0.1232 in.
			3.14 – 3.16 mm	0.1236 – 0.1244 in.
			3.17 – 3.19 mm	0.1248 – 0.1256 in.
			3.20 – 3.22 mm	0.1260 – 0.1268 in.
	Drive pinion adjusting washer thickness		2.26 – 2.28 mm	0.0890 – 0.0898 in.
			2.29 – 2.31 mm	0.0902 – 0.0909 in.

Specifications (Cont'd)

Differential (cont'd)	Drive pinion adjusting washer thickness (cont'd)	2.32 – 2.34 mm	0.0913 – 0.0921 in.
		2.35 – 2.37 mm	0.0925 – 0.0933 in.
		2.38 – 2.40 mm	0.0937 – 0.0945 in.
		2.41 – 2.43 mm	0.0949 – 0.0957 in.
		2.44 – 2.46 mm	0.0961 – 0.0969 in.
		2.47 – 2.49 mm	0.0972 – 0.0980 in.
		2.50 – 2.52 mm	0.0984 – 0.0992 in.
		2.53 – 2.55 mm	0.0996 – 0.1004 in.
		2.56 – 2.58 mm	0.1008 – 0.1016 in.
		2.59 – 2.61 mm	0.1020 – 0.1028 in.
		2.62 – 2.64 mm	0.1031 – 0.1039 in.
		2.65 – 2.67 mm	0.1043 – 0.1051 in.
		2.68 – 2.70 mm	0.1055 – 0.1063 in.

Torque Specifications

Part tightened		kg-cm	ft-lb	N-m
FRONT AXLE AND SUSPENSION				
Tie rod end x Steering knuckle		500	36	49
Ball joint x Lower arm		1,300	94	127
Hub bearing lock nut		1,900	137	186
Transmission case protector bolt		185	13	18
Drive shaft x Side gear shaft		660	48	65
REAR AXLE AND SUSPENSION				
Brake caliper x Axle carrier		475	34	47
Strut rod x Axle carrier		1,150	83	113
Rear speed sensor x Axle carrier (w/ ABS)		195	14	19
Axle carrier x Backing plate		730	53	72
Parking brake cable x Backing plate		80	69 in.-lb	7.8
Rear drive shaft x Axle hub		1,900	137	186
Suspension arm x Body		1,150	83	113
Suspension arm x Axle carrier		1,250	90	123
Strut rod x Body		1,150	83	113
Hub bolt x Wheel		1,050	76	103
Drive shaft x Side gear shaft		700	51	69
Propeller shaft x Companion flange		750	54	74
Rear crossmember x Body		730	53	72
Differential case x Ring gear		985	71	97
Bearing cap x Differential carrier		800	58	78
Differential carrier x Carrier cover		475	34	47
Differential x Support member	Upper side	970	70	95
	Rear side	1,500	108	147
Drain plug		500	36	49
Filler plug		400	29	39
Shock absorber x Axle carrier		2,600	188	255
Shock absorber x Stabilizer link		650	47	64
Stabilizer bracket x Body		195	14	19

BRAKE SYSTEM**Specifications**

Front brake	Disc thickness	STD	25.0 mm	0.984 in.
		Limit	23.0 mm	0.906 in.

Torque Specifications

Part tightened	kg-cm	ft-lb	N·m
ABS actuator x Proportioning valve bracket	90	78 in.-lb	8.8
ABS actuator x 3-way union	160	12	16
Proportioning valve installation bolt	90	78 in.-lb	8.8
ABS actuator x Actuator bracket	55	48 in.-lb	5.4
Front speed sensor installation bolt	80	69 in.-lb	7.8
Rear speed sensor installation bolt	195	14	19

STEERING**Specifications**

Power steering	Oil pressure at Idle speed	80 – 85 kg/cm ² (1,138 – 1,209 psi, 7,845 – 8,336 kPa)
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Torque Specifications

	Part tightened	kg-cm	ft-lb	N·m
Power steering (Power steering pump)	Drive pulley nut	440	32	43
	Pressure tube union bolt	525	38	51
	Pressure tube union nut	370	27	36
	PS pump x Bracket	440	32	43
	Upper	400	29	39
	Lower	500	36	49
	Tie rod end x Knuckle arm	1,550	112	152
	Lower crossmember RH and LH sides	400	29	39
(Gear housing)	Gear housing x Body	600	43	59
	Gear housing x Return line	450	33	44
	Gear housing x Pressure line	450	33	44
	Center member x Body	530	38	52
	Tie rod end x Knuckle arm	500	36	49
	Control valve shaft x Universal joint	360	26	35
	Lower crossmember RH and LH sides	1,550	112	152
	Center	400	29	39
	Tube clamp set bolt	130	9	13
	Engine x Engine mounting	790	57	77
	Engine mounting x Insulator	890	64	87
	Engine mounting x Center member	740	54	73
	Exhaust front pipe x Manifold	630	46	62
	Exhaust front pipe x Center pipe	440	32	43
	Propeller shaft x Intermediate shaft	750	54	74
	Stabilizer bar x Link	360	26	35
	Stabilizer bar bracket	180	13	18

LUBRICANT

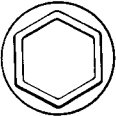
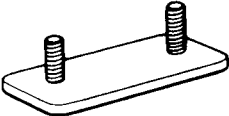
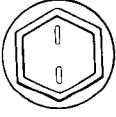
Item	Capacity			Classification
	Liters	US qts	Imp. qts	
Engine oil				API grade SG, multigrade fuel-efficient and recommended viscosity oil
Drain and refill				
w/ Oil filter change	3.9	4.1	3.4	
w/o Oil filter change	3.6	3.8	3.2	
Dry fill	4.3	4.5	3.8	
Manual transaxle oil (E150F)	5.2	5.5	4.6	API GL-5 Viscosity: Above – 18°C (0°F) SAE90 Below – 18°C (0°F) SAE80W
Differential oil	1.1	1.2	1.0	API GL-5 Hypoid gear oil Viscosity: Above – 18°C (0°F) SAE90 Below – 18°C (0°F) SAE80W-90
Brake fluid	–			SAE J1703 or FMVSS No. 116 DOT3

STANDARD BOLT TORQUE SPECIFICATIONS

	Page
STANDARD BOLT TORQUE SPECIFICATIONS	B-2

STANDARD BOLT TORQUE SPECIFICATIONS

HOW TO DETERMINE BOLT STRENGTH

	Mark	Class		Mark	Class
Hexagon head bolt	4 — 5 — 6 — Bolt head No. 7 — 8 — 9 — 10 — 11 —	4T 5T 6T 7T 8T 9T 10T 11T	Stud bolt	 No mark	4T
	 No mark	4T		 Grooved	6T
Hexagon flange bolt w/ washer hexagon bolt	 No mark	4T	Welded bolt		4T
Hexagon head bolt	 Two protruding lines	5T			
Hexagon flange bolt w/ washer hexagon bolt	 Two protruding lines	6T			
Hexagon head bolt	 Three protruding lines	7T			
Hexagon head bolt	 Four protruding lines	8T			

SPECIFIED TORQUE FOR STANDARD BOLTS

Class	Diameter mm	Pitch mm	Specified torque					
			Hexagon head bolt			Hexagon flange bolt		
			kg-cm	ft-lb	N·m	kg-cm	ft-lb	N·m
4T	6	1	55	48 in.-lb	5	60	52 in.-lb	6
	8	1.25	130	9	12.5	145	10	14
	10	1.25	260	19	26	290	21	29
	12	1.25	480	35	47	540	39	53
	14	1.5	760	55	74	850	61	84
	16	1.5	1,150	83	115	—	—	—
5T	6	1	65	56 in.-lb	6.5	75	65 in.-lb	7.5
	8	1.25	160	12	15.5	175	13	17.5
	10	1.25	330	24	32	360	26	36
	12	1.25	600	43	59	670	48	65
	14	1.5	930	67	91	1,050	76	100
	16	1.5	1,400	101	140	—	—	—
6T	6	1	80	69 in.-lb	8	90	78 in.-lb	9
	8	1.25	195	14	19	210	15	21
	10	1.25	400	29	39	440	32	44
	12	1.25	730	53	71	810	59	80
	14	1.5	1,100	80	110	1,250	90	125
	16	1.5	1,750	127	170	—	—	—
7T	6	1	110	8	10.5	120	9	12
	8	1.25	260	19	25	290	21	28
	10	1.25	530	38	52	590	43	58
	12	1.25	970	70	95	1,050	76	105
	14	1.5	1,500	108	145	1,700	123	165
	16	1.5	2,300	166	230	—	—	—
8T	8	1.25	300	22	29	330	24	33
	10	1.25	620	45	61	690	50	68
	12	1.25	1,100	80	110	1,250	90	120
9T	8	1.25	340	25	34	380	27	37
	10	1.25	710	51	70	790	57	78
	12	1.25	1,300	94	125	1,450	105	140
10T	8	1.25	390	28	38	430	31	42
	10	1.25	800	58	78	890	64	88
	12	1.25	1,450	105	140	1,600	116	155
11T	8	1.25	430	31	42	480	35	47
	10	1.25	890	64	87	990	72	97
	12	1.25	1,600	116	155	1,800	130	175

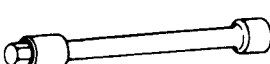
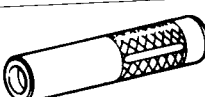
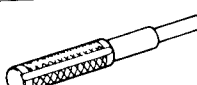
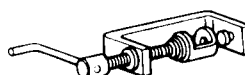
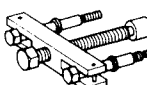
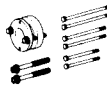
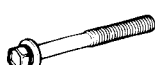
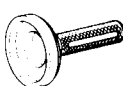
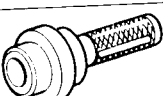
SST AND SSM

REFER TO 1990 CELICA REPAIR MANUAL (Pub. No. RM149U)

NOTE: The following pages contain only the points which differ from the above listed manual.

	Page
SST (SPECIAL SERVICE TOOLS)	C-2
SSM (SPECIAL SERVICE MATERIALS)	C-12

SST (SPECIAL SERVICE TOOLS)

Section			EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR
Part Name	Part No.	Illustration									Front	Rear		
	09043-38100	Hexagon 10 mm Wrench	●						●					
	09043-88010	Hexagon 8 mm Wrench									●			
	09201-41020	Valve Stem Oil Seal Replacer	●											
	09201-60011	Valve Guide Bushing Remover & Replacer							● ^{*1}					
	09201-70010	Valve Guide Bushing Remover & Replacer	●											
	09202-70010	Valve Spring Compressor	●											
	09213-31021	Crankshaft Pulley Puller	●											
	09213-54015	Crankshaft Pulley Holding Tool	●											
	(90119-08216)	(Bolt)	●											
	09222-30010	Connecting Rod Bushing Remover & Replacer	●											
	09223-15010	Crankshaft Rear Oil Seal Replacer							● ^{*2}		●			
	09223-50010	Crankshaft Front Oil Seal Replacer	● ^{*3}											

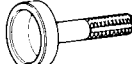
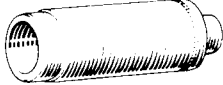
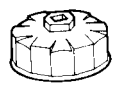
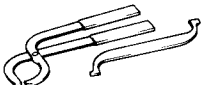
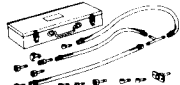
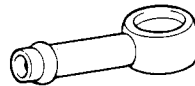
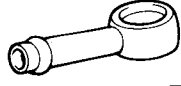
Remarks:

*1 Speedometer driven gear oil seal

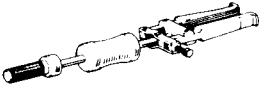
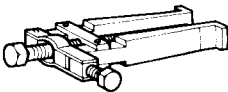
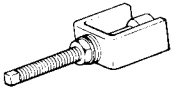
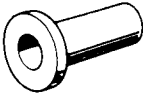
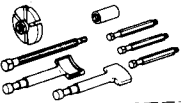
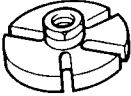
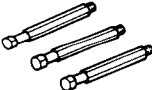
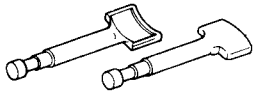
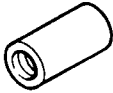
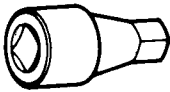
*2 Side bearing oil seal

*3 Camshaft oil seal

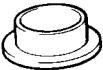
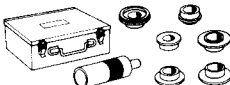
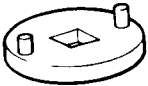
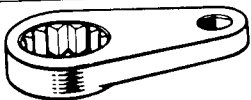
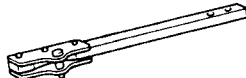
SST (SPECIAL SERVICE TOOLS) (Cont'd)

Section			EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR
Part Name											Front	Rear		
Part No.														
Illustration														
	09223-63010	Crankshaft Rear Oil Seal Replacer	●											
	09226-10010	Crankshaft Front & Rear Bearing Replacer	●											
	09228-06500	Oil Filter Wrench				●								
	09240-00020	Wire Gauge Set					●							
	09248-55010	Valve Clearance Adjust Tool Set	●											
	09249-63010	Torque Wrench Adaptor	●											
	09268-41045	Injection Measuring Tool Set			●									
	(09268-41080)	(No. 6 Union)			●									
	(90405-09015)	(No. 1 Union)			●									
	09268-45012	EFI Fuel Pressure Gauge			●									
	09268-74010	Injector Remover			●									
	09301-17010	Clutch Guide Tool						●						

SST (SPECIAL SERVICE TOOLS) (Cont'd)

Section			EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR
Part Name	Part No.	Illustration									Front	Rear		
	09307-12010	Extension Housing Bushing Replacer							●					
	09308-00010	Oil Seal Puller							●		●	●		
	09308-10010	Oil Seal Puller										●		
	09309-32050	5th Driven Gear Replacer							●					
	09309-36010	Transmission Rear Bearing Replacer							●			●		
	09310-17010	Transaxle Gear Remover & Replacer							●					
	(09310-07010)	(Plate)							●					
	(09310-07020)	(Center Bolt)							●					
	(09310-07030)	(Set Bolt)							●					
	(09310-07040)	(Claw)							●					
	(09310-07050)	(Adaptor)							●					
	09313-30021	Detent Ball Plug Socket							●					

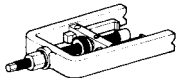
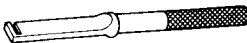
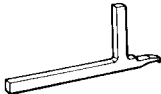
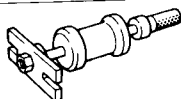
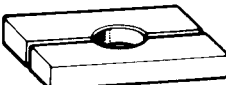
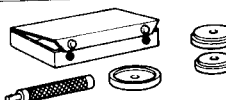
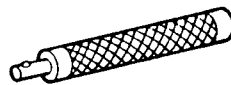
SST (SPECIAL SERVICE TOOLS) (Cont'd)

Section			EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR
Part Name	Part No.	Illustration									Front	Rear		
	09316-20011	Transfer Bearing Replacer							●					
	09316-60010	Transmission & Transfer Bearing Replacer							●					
	(09316-00010)	(Replacer Pipe)							●					
	(09316-00020)	(Replacer "A")							●					
	(09316-00040)	(Replacer "C")							●					
	(09316-00050)	(Replacer "D")							●					
	(09316-00060)	(Replacer "E")							●					
	(09316-00070)	(Replacer "F")							●					
	09318-20010	Transfer Side Bearing Adjusting Nut Wrench							●					
	09325-20010	Transmission Oil Plug							●	●				●
	09326-20011	Output Shaft Bearing Lock Nut Wrench							●					
	09330-00021	Companion Flange Holding Tool	●								●		●	

Remark:

*1 Crankshaft pulley

SST (SPECIAL SERVICE TOOLS) (Cont'd)

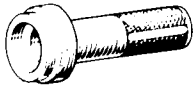
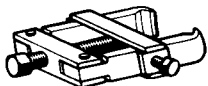
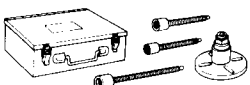
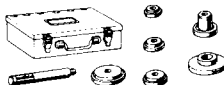
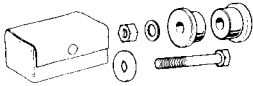
Section			EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR
Part Name	Part No.	Illustration									Front	Rear		
	09332-25010	Universal Joint Bearing Remover & Replacer												
	09333-00013	Clutch Diaphragm Spring Aligner												
	09504-22011	Differential Side Washer Remover & Replacer												
	09506-30012	Differential Drive Pinion Bearing Cone Replacer												
	09506-35010	Differential Drive Pinion Rear Bearing Replacer												
	09520-24010	Differential Side Gear Shaft Puller												
	09527-21011	Rear Axle Shaft Bearing Remover												
	09550-10012	Replacer Set "B"												
	(09252-10010)	(No. 1 Replacer) Handle												
	(09555-10010)	(Differential Drive Pinion Rear Bearing) Replacer												
	09550-22011	Rear Axle Bearing & Differential Tool Set												
	(09550-00020)	(Handle)												

Remarks:

*1 Output shaft rear bearing

*2 No.2 clutch hub and input shaft rear bearing

SST (SPECIAL SERVICE TOOLS) (Cont'd)

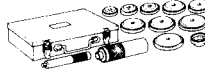
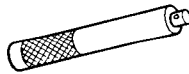
Section	Part Name	Part No.	Illustration	EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR
												Front	Rear		
		(09550-00031)	(Replacer)										●		
		09554-22010	Differential Oil Seal Replacer										●		
		09556-22010	Drive Pinion Front Bearing Remover										●		
		09557-22022	Companion Flange Remover & Replacer								●		●		
		(09557-22040)	(No. 2 Bolt)										●		
		(09557-22050)	(No. 3 Bolt)								●				
		09608-12010	Front Hub & Drive Pinion Bearing Replacer Set							●					
		(09608-00020)	(Remover & Replacer) Handle							● ^{*1}					
		(09608-00060)	(Drive Pinion Front Bearing Cup Replacer)							● ^{*2}					
		(09608-00080)	(Differential Side Bearing Cone Replacer)							● ^{*1}					
		09608-16041	Front Hub Bearing Adjusting Tool									●	●		
		(09608-02020)	(Bolt & Nut)									●	●		

Remarks:

*1 Input shaft front seal

*2 Input shaft bearing

SST (SPECIAL SERVICE TOOLS) (Cont'd)

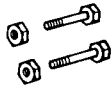
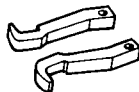
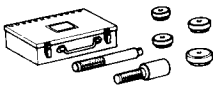
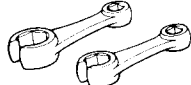
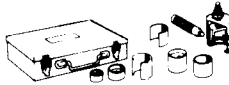
Section			EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR										
Part Name											Front	Rear												
Part No.																								
Illustration																								
	(09608-02040)	(Retainer)										●	●											
	09608-30012	Front Hub & Drive Pinion Bearing Tool Set											●											
	(09608-00060)	(Drive Pinion Front Bearing Cup Replacer)											●											
	(09608-04020)	(Handle)											●											
	(09608-04100)	(Drive Pinion Front Bearing Cup Replacer)											●											
	(09608-04110)	(Drive Pinion Rear Bearing Cup Replacer)											●											
	09608-32010	Steering Knuckle Oil Seal Replacer											●											
	09608-35014	Axle Hub & Drive Pinion Bearing Tool Set							●															
	(09608-06020)	(Handle)							*1															
	(09608-06120)	(Front Hub Outer Bearing Cup Replacer)							●															
	(09608-06180)	(Drive Pinion Rear Bearing Cup Replacer)							*2															
	09611-22012	Tie Rod End Puller													●									

Remarks:

*1 Ring gear mounting case side bearing outer race

*2 Front bearing outer race and rear bearing outer race

SST (SPECIAL SERVICE TOOLS) (Cont'd)

Section			EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR
Part Name											Front	Rear		
Part No.														
Illustration														
	09612-65014	Steering Worm Bearing Puller							●					
	(09612-01050)	(Hanger Pin with Nut)							*1					
	(09612-01060)	(Claw "E")							*1					
	09616-30011	Steering Worm Bearing Adjusting Screw Wrench	*2											
	09616-30020	Steering Worm Bearing Adjusting Screw Wrench												●
	09620-30010	Steering Gear Box Replacer Set							●					
	(09627-30010)	(Steering Sector Shaft Bushing Replacer)							*3					
	(09631-00020)	(Handle)							*3					
	09628-62011	Ball Joint Puller									●			
	09631-22020	Power Steering Hose Nut 14 x 17 mm Wrench Set			*4									●
	09636-20010	Upper Ball Joint Dust Cover Replacer											●	
	09710-22020	Front Suspension Bushing Tool Set											●	

Remarks:

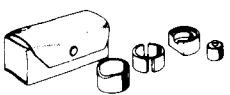
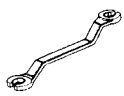
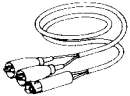
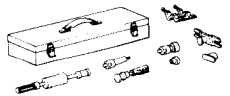
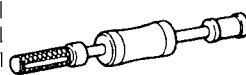
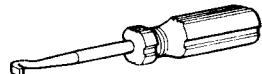
*1 Input shaft bearing

*2 Oil pump pulley

*3 Control shaft cover oil seal, differential lock shift lever shaft oil seal and shift fork shaft oil seal

*4 Fuel line flare nut

SST (SPECIAL SERVICE TOOLS) (Cont'd)

Section			EM	TC	FI	LU	IG	CL	MT	PR	SA		BR	SR										
Part Name											Front	Rear												
Part No.																								
Illustration																								
	(09710-01030)	(Upper Arm Bushing Replacer)											●											
	09726-10010	Lower Suspension Arm Bushing Remover & Replacer										●												
	(09726-00030)	(Spacer)										●												
	09751-36011	Brake Tube Union Nut 10 x 12 mm Wrench												●										
	09816-30010	Oil Pressure Switch Socket	*1			●																		
	09842-30050	Wire "A" EFI Inspection			●																			
	09842-30060	Wire "E" EFI Inspection			●																			
	09843-18020	Diagnosis Check Wire			●		●							●										
	09910-00015	Puller Set							●															
	(09911-00011)	(Puller Clamp)							●															
	(09912-00010)	(Puller Slide Hammer)							●															
	09921-00010	Spring Tension Tool							*2															

Remarks:

*1 Knock sensor

*2 Speedometer driven gear oil seal

SST (SPECIAL SERVICE TOOLS) (Cont'd)

[illegible]

SSM (SPECIAL SERVICE MATERIALS)

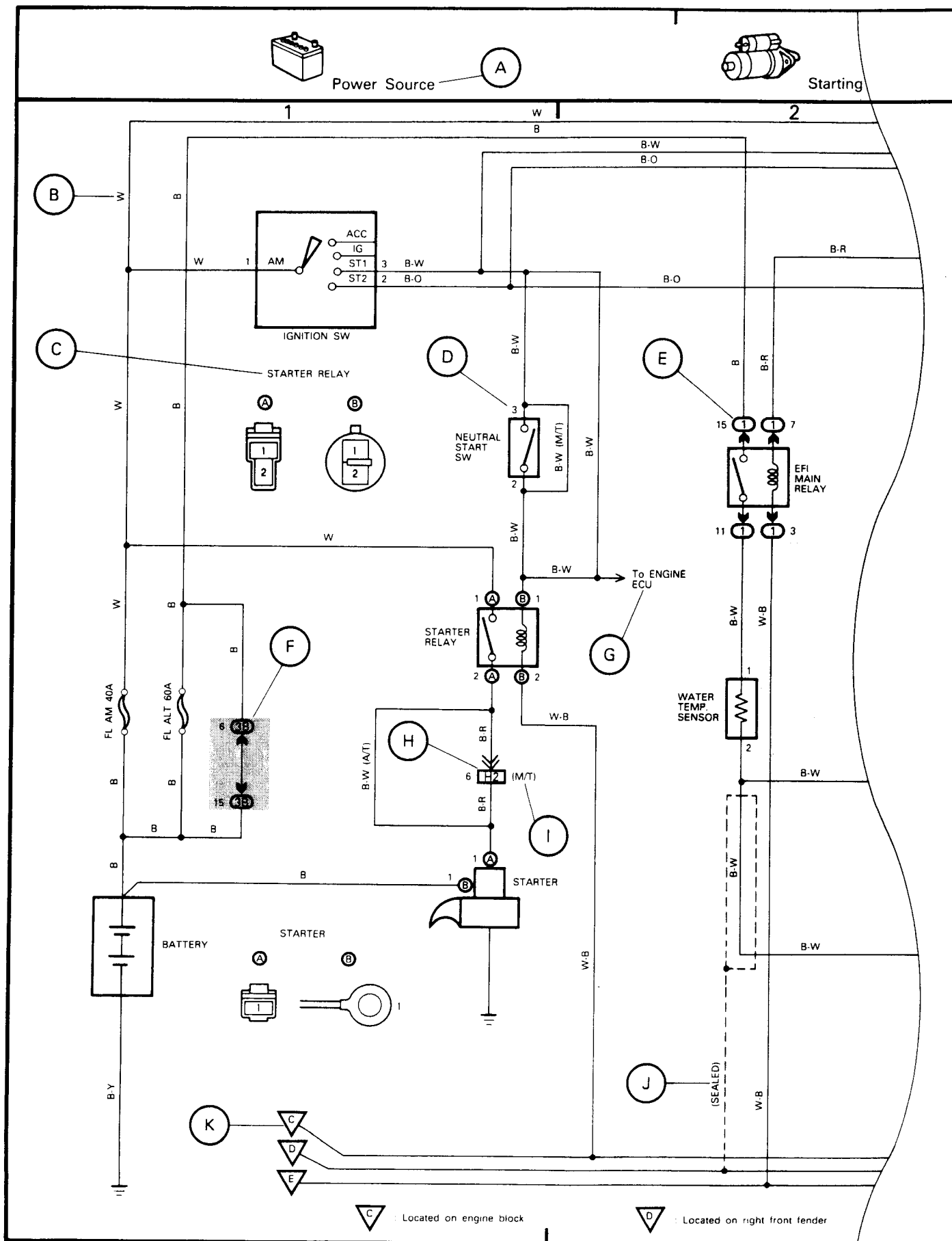
Part Name	Part No.	Sec.	Use etc.
Seal packing or equivalent	08826-00080	EM	Camshaft bearing cap
Seal packing 1281, Three bond 1281 or equivalent	08826-00090	MT	Transaxle case x Transmission case Transmission case x Case cover Transfer left case Transmission x Transfer Transfer inspection cover
		SA	Differential carrier cover
Seal packing 1121, Three bond 1121 or equivalent	08826-00801	PR	Universal joint cover
		SA	Inboard joint cover
Adhesive 1324, Three bond 1324 or equivalent	08833-00070	EM	Flywheel mount bolt for M/T
		EC	BVSV for EVAP
Adhesive 1344, Three bond 1344, Loctite 242 or equivalent	08833-00080	EM	No.1 idler pulley pivot bolt
		LU	Oil pressure switch
		MT	Rear bearing retainer set screw Reverse restrict pin holder plug Straight screw plug Control shaft cover Bellcrank support lock bolt Transfer set bolt Transfer left case bolt

ELECTRICAL WIRING DIAGRAMS

ABBREVIATION

The following abbreviations are used in this wiring diagram.

ABS	=	Anti-Lock Brake System
A/C	=	Air Conditioner
CB	=	Circuit Breaker
CD	=	Compact Disc
COMB.	=	Combination
ECU	=	Electronic Control Unit
EFI	=	Electronic Fuel Injection
FL	=	Fusible Link
J/B	=	Junction Block
LH	=	Left-Hand
R/B	=	Relay Block
RH	=	Right-Hand
SRS	=	Supplemental Restraint System
SW	=	Switch
TEMP.	=	Temperature
VSV	=	Vacuum Switching Valve
w	=	With
w/o	=	Without



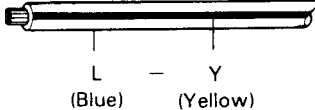
Ⓐ: System Title

Ⓑ: Indicates the wiring color.

Wire colors are indicated by an alphabetical code.

B = Black L = Blue R = Red
 BR = Brown LG = Light Green V = Violet
 G = Green O = Orange W = White
 GR = Gray P = Pink Y = Yellow

The first letter indicates the basic wire color and the second letter indicates the color of the stripe.

Example: L—Y 

Ⓒ: Indicates the connector to be connected to a part (the numeral indicates the pin No.)

Ⓓ: Indicates the pin number of the connector. The numbering system is different for female and male connectors.

Example: Numbered in order from upper left to lower right Numbered in order from upper right to lower left



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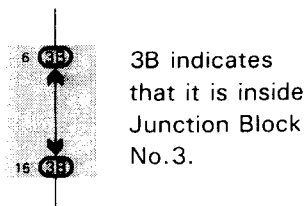
The numbering system for the overall wiring diagram is the same as above.

Ⓔ: Indicates a Relay Block. No shading is used and only the Relay Block No. is shown to distinguish it from the J/B.

Example: ① Indicates Relay Block No.1.

Ⓕ: Junction Block (The number in the circle is the J/B No. and the connector code is shown beside it). Junction Blocks are shaded to clearly separate them from other parts (different junction blocks are shaded differently for further clarification).

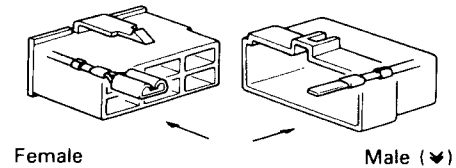
Example:



Ⓖ: Indicates related system.

Ⓗ: Indicates the wiring harness and wiring harness connector. The wiring harness with male terminal is shown with arrows (↗). Outside numerals are pin numbers.

- All connectors are shown from the open end, and the lock is on top.



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Ⓘ: () are used to indicate different wiring and connector, etc. when the vehicle model, engine type, or specification is different.

Ⓙ: Indicates a sealed wiring harness.

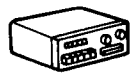
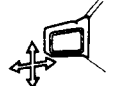
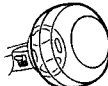
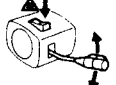
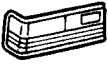


Ⓚ: Indicates a ground point.

SYSTEM INDEX

CELICA

1990 Model (Page 1 to Page 22)

SYSTEMS	LOCATION	SYSTEMS	LOCATION
Air Conditioner, Cooler and Heater	 14-4 15-2 (Manual) 17-4 18-2 (Auto)	Power Seats	 5-3
ABS (Anti-lock Brake System)	 12-3	Power Source	 1~17 18-5
Auto Antenna	 10-3	Power Windows	 8-3
Auto Tilt Away Steering	 10-2	Radiator Fan and Condenser Fan	 15-3
Back-up Lights	 8-2	Radio and Player	 10-4 (w/o CD) 11-3 (w/ CD)
Charging	 1-3	Rear Window Defogger	 9-4
Cigarette Lighter	 4-4	Rear Wiper and Washer	 6-3
Clock	 4-4	Remote Control Mirrors	 11-2
Combination Meter	 16-3	SRS Airbag	 6-4
Cruise Control	 18-6	Starting	 1-2
Door Locks	 9-2	Stop Lights	 5-4
Engine Control	 13-4	Sun Roof	 8-4
Fog lights	 1-4	Taillights	 3-2
Front Wiper and Washer	 6-2	Turn Signal and Hazard	 5-2
Headlights	 2-3	Unlock and Seat Belt Warning	 7-2
Horn	 5-3		
Illumination	 4-2	Junction Block and Wire Harness Connector	19-2
Interior Lights	 7-3	Connector Joining Wire Harness and Wire Harness	20-2 21-2 22-2
Light Reminder	 3-4		



Power Source



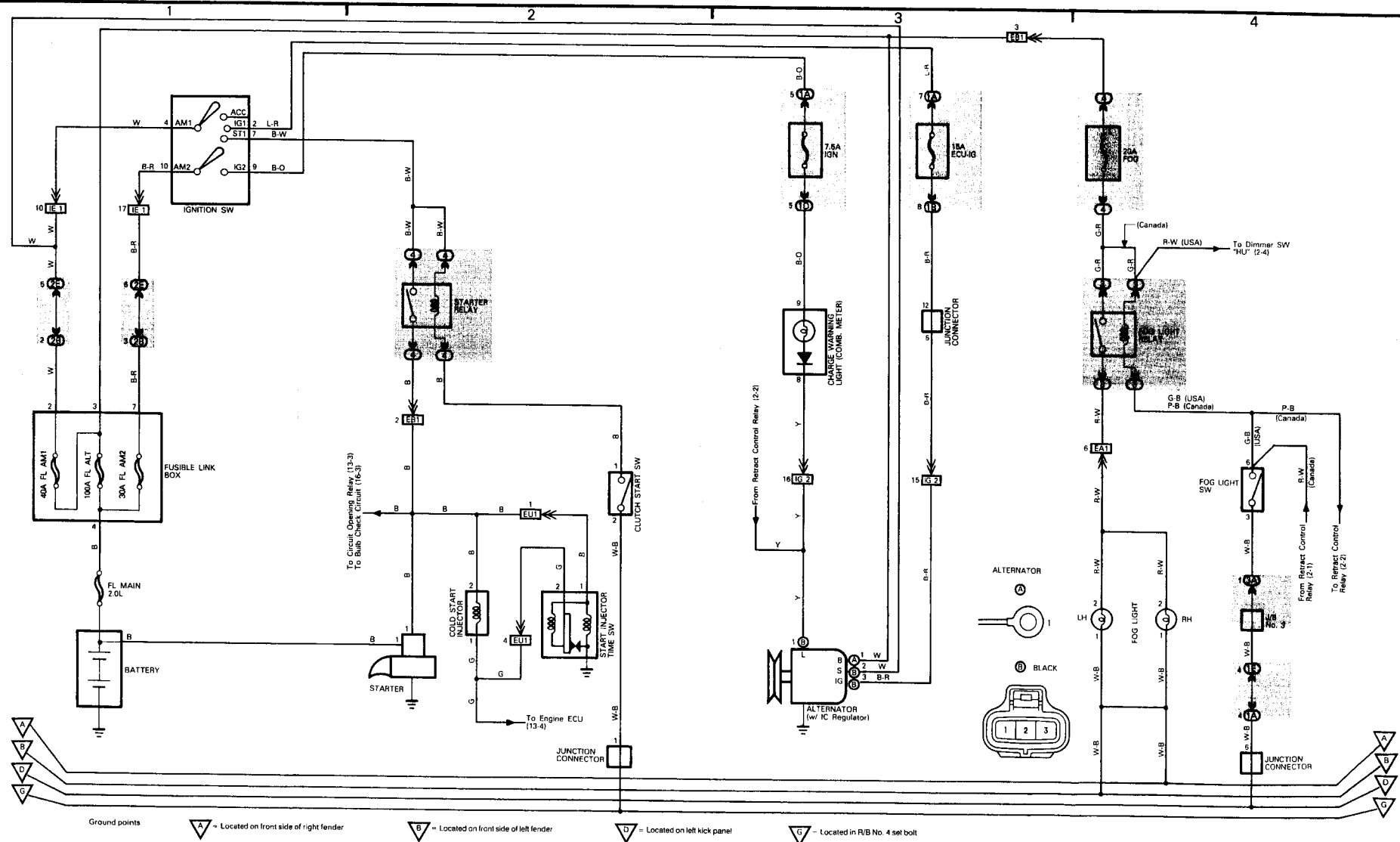
Starter



Charging



Fog Light







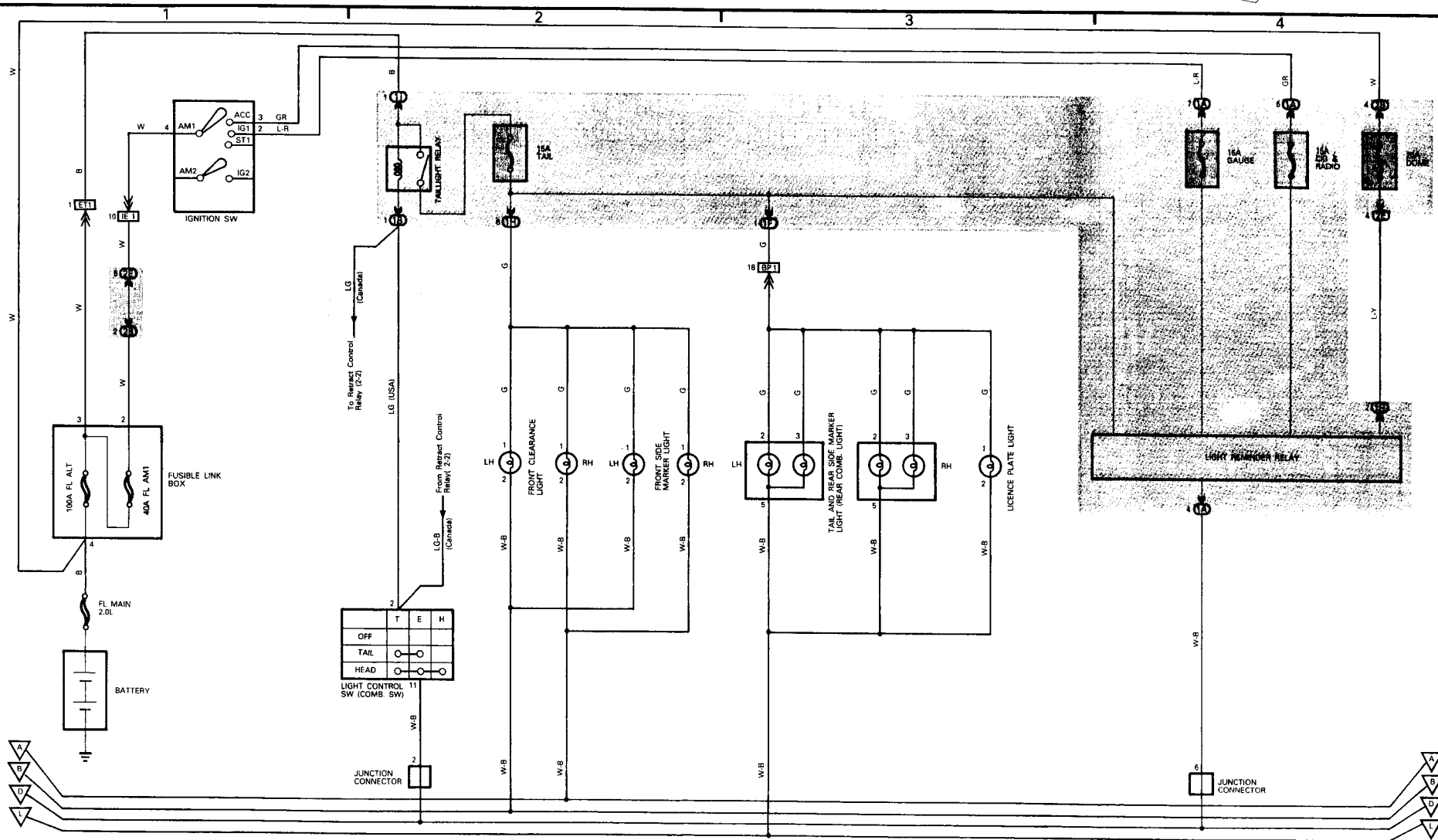
Power Source



Taillight



Light Reminder



Ground points

△ = Located on front side of right fender

△ = Located on front side of left fender

△ = Located on left kick panel

△ = Located on center of back panel



Power Source



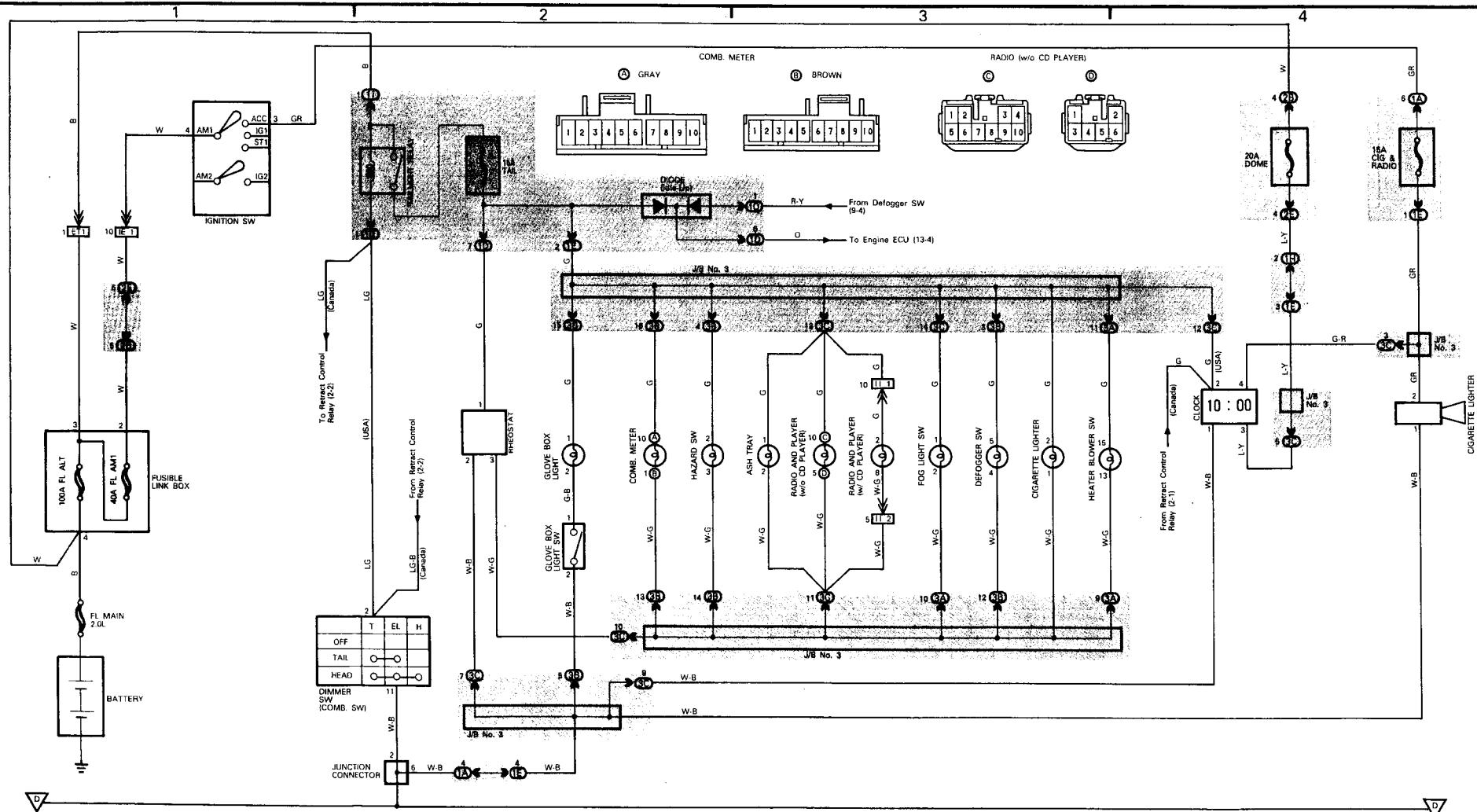
Illumination



Clock



Cigarette Lighter



Ground points

D = Located on left kick panel



Power Source



Turn Signal and Hazard



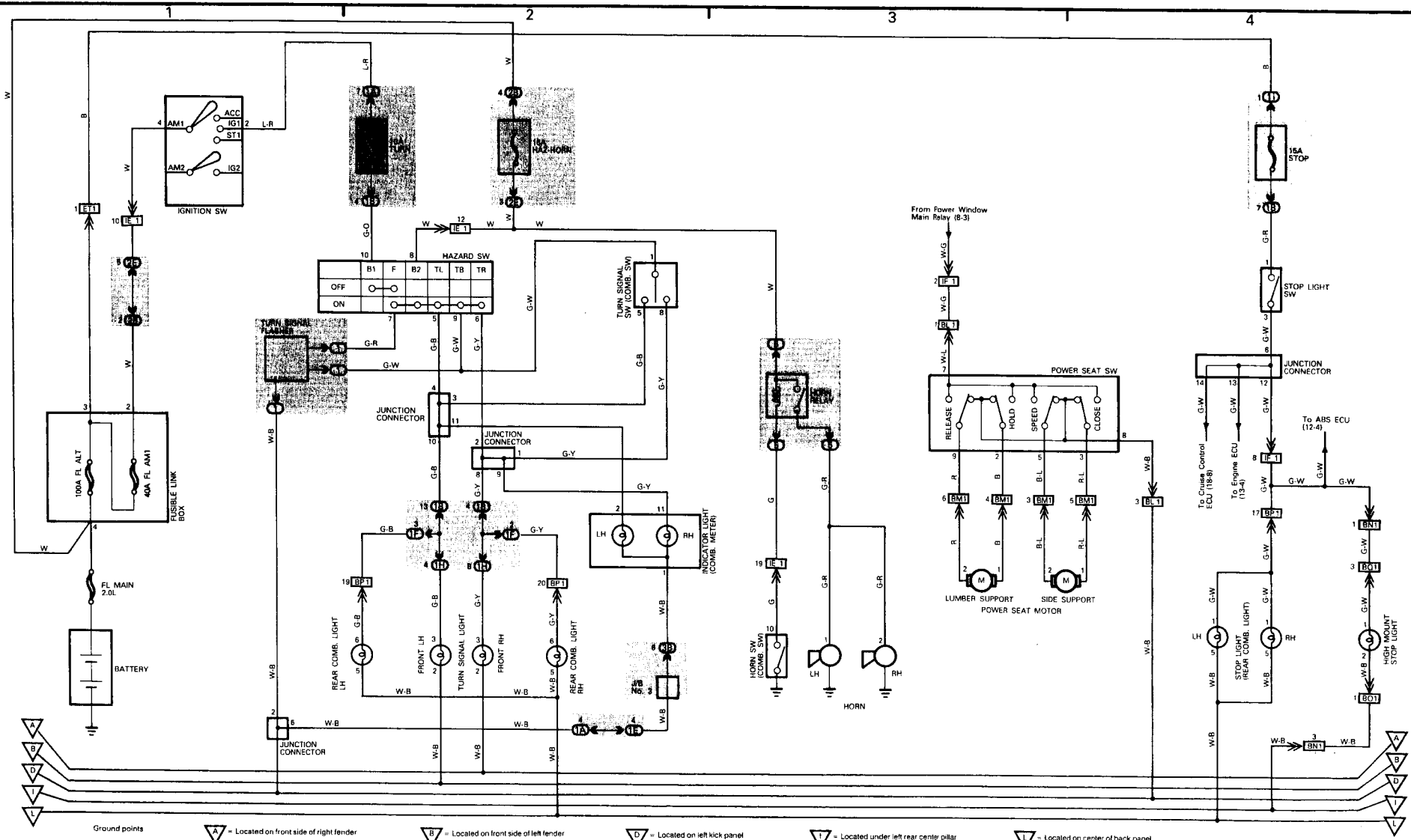
Horn



Power Seat



Stop Light





Power Source



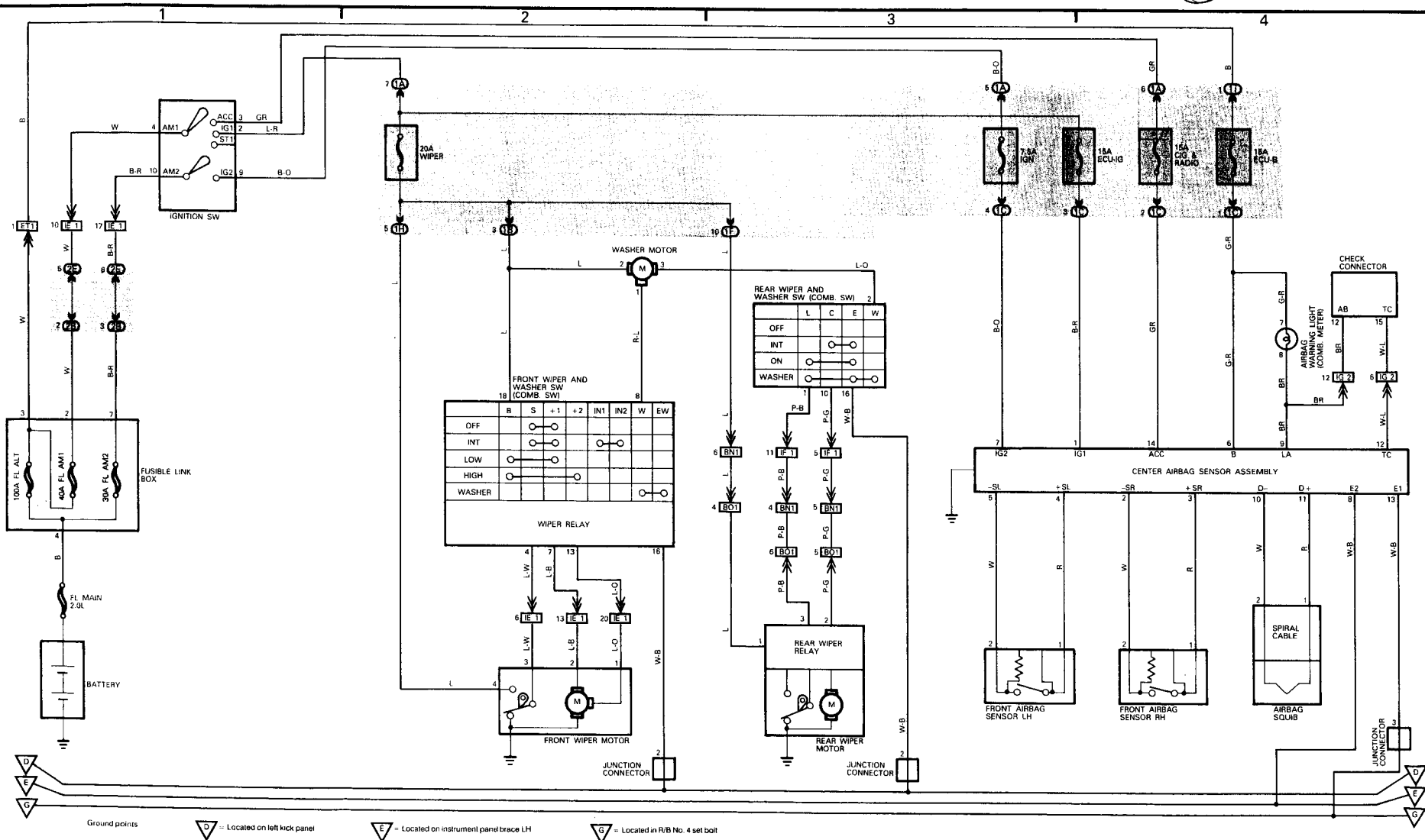
Front Wiper and Washer

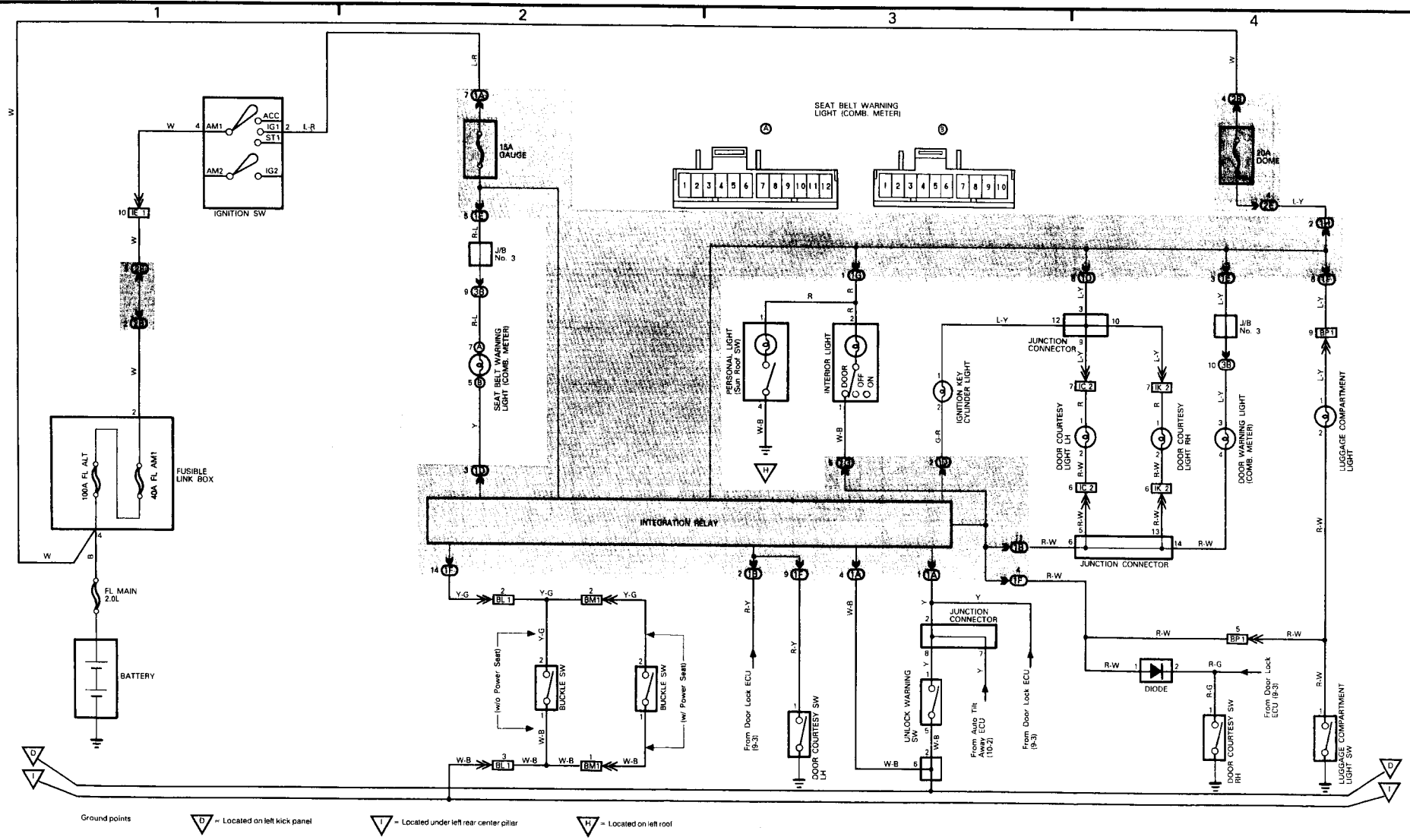
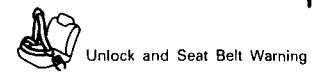


Rear Wiper and Washer



SRS Airbag



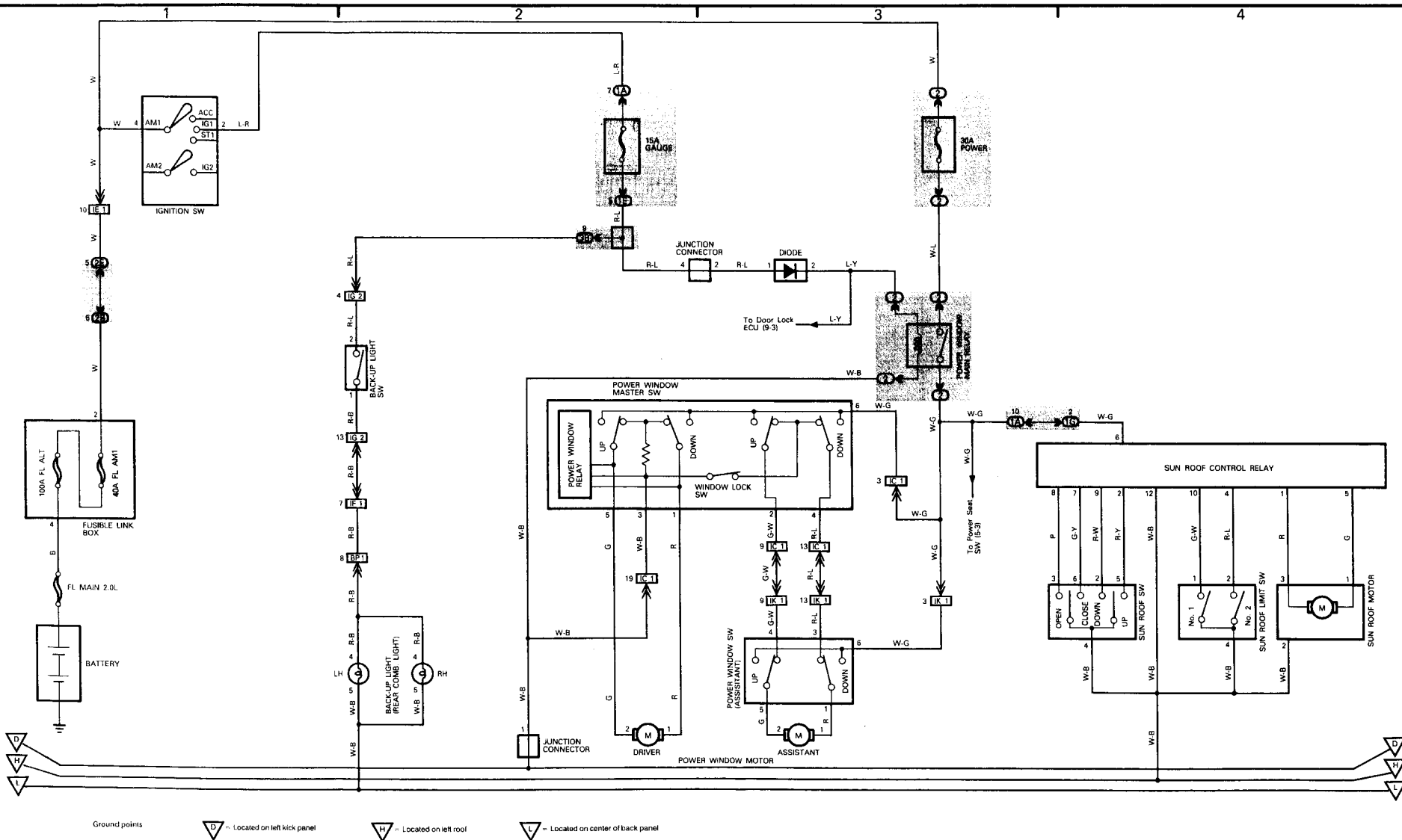


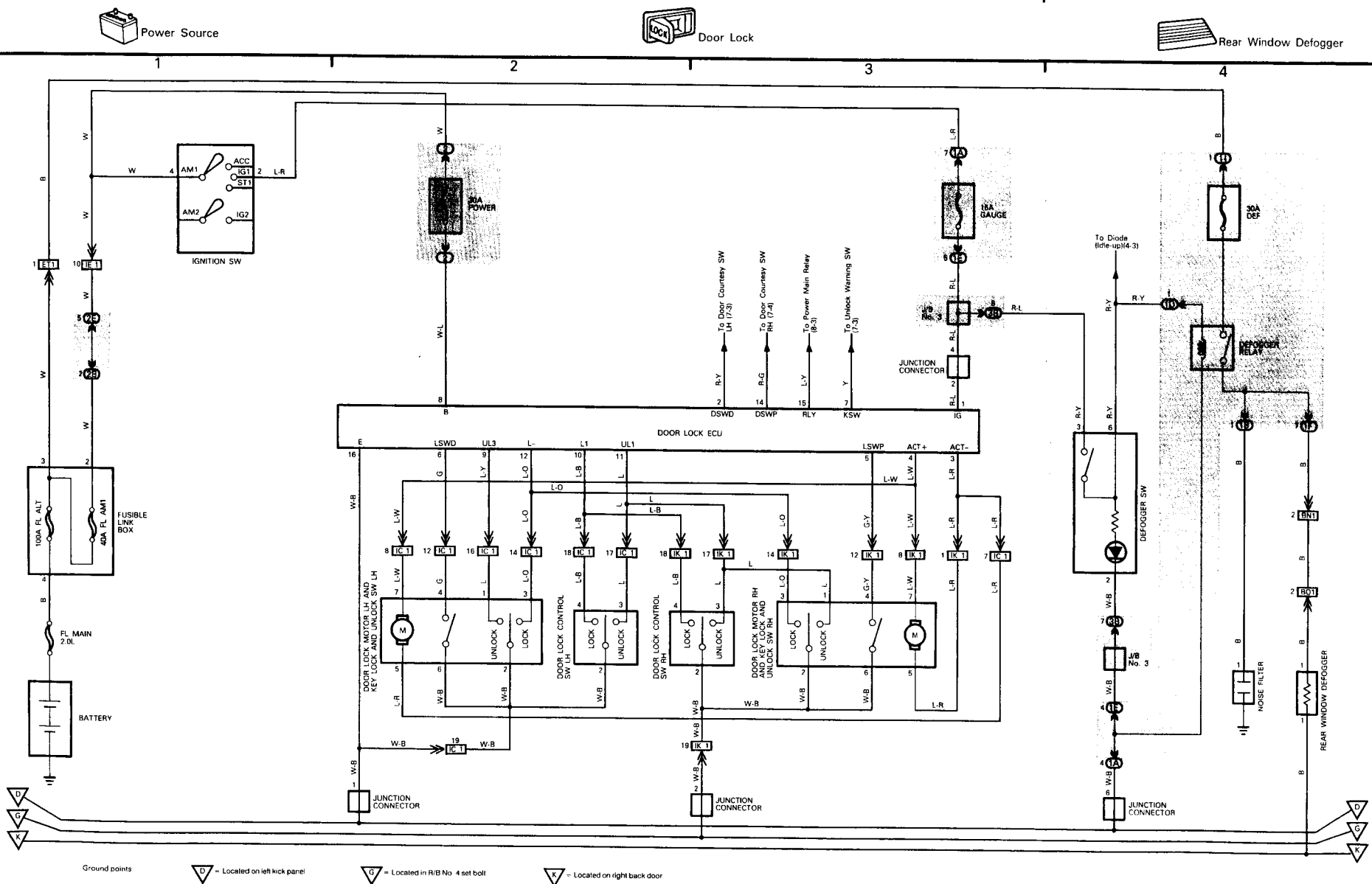
Ground points

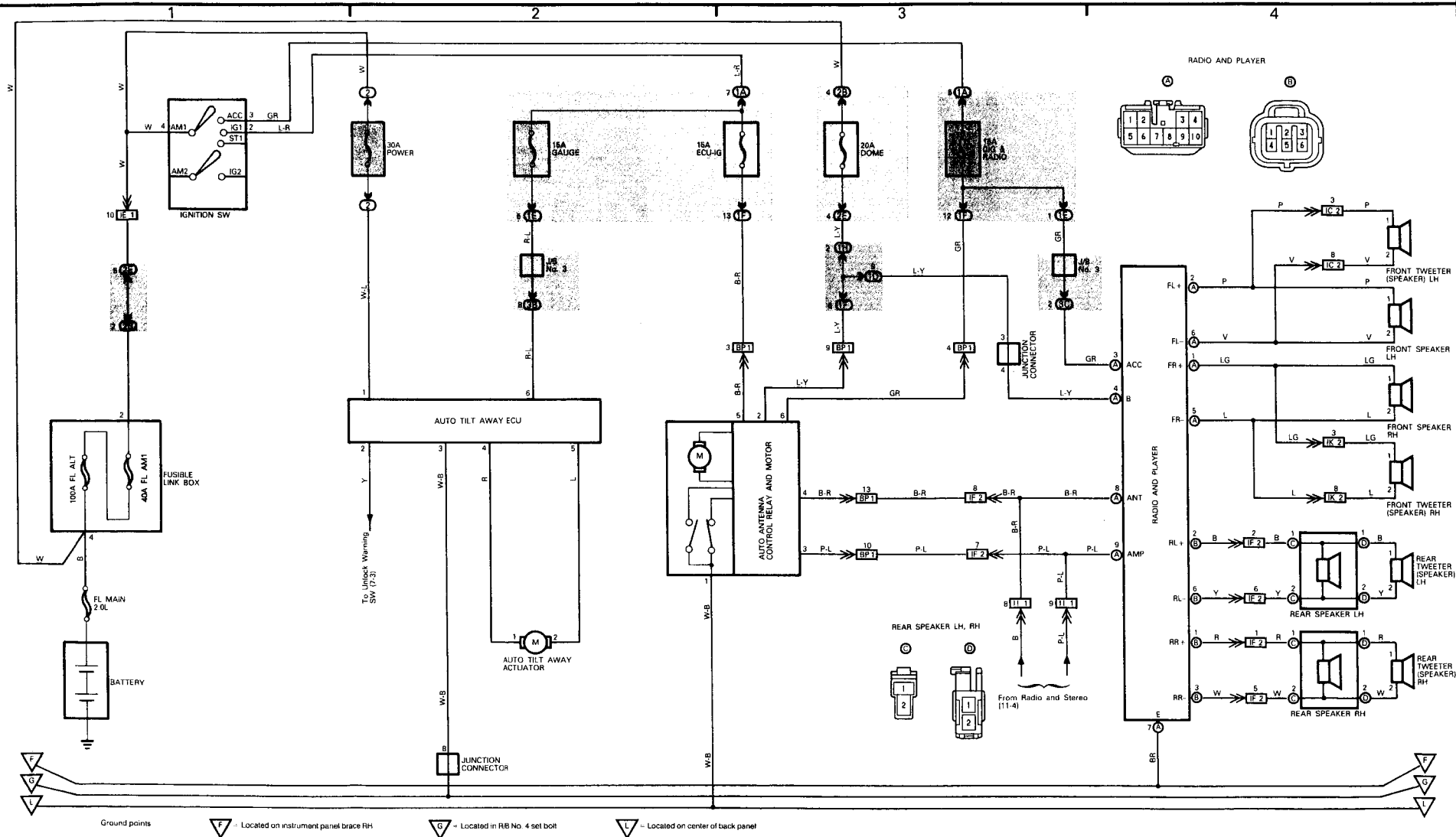
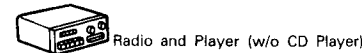
D = Located on left kick panel

V = Located under left rear center pillar

H = Located on left roof









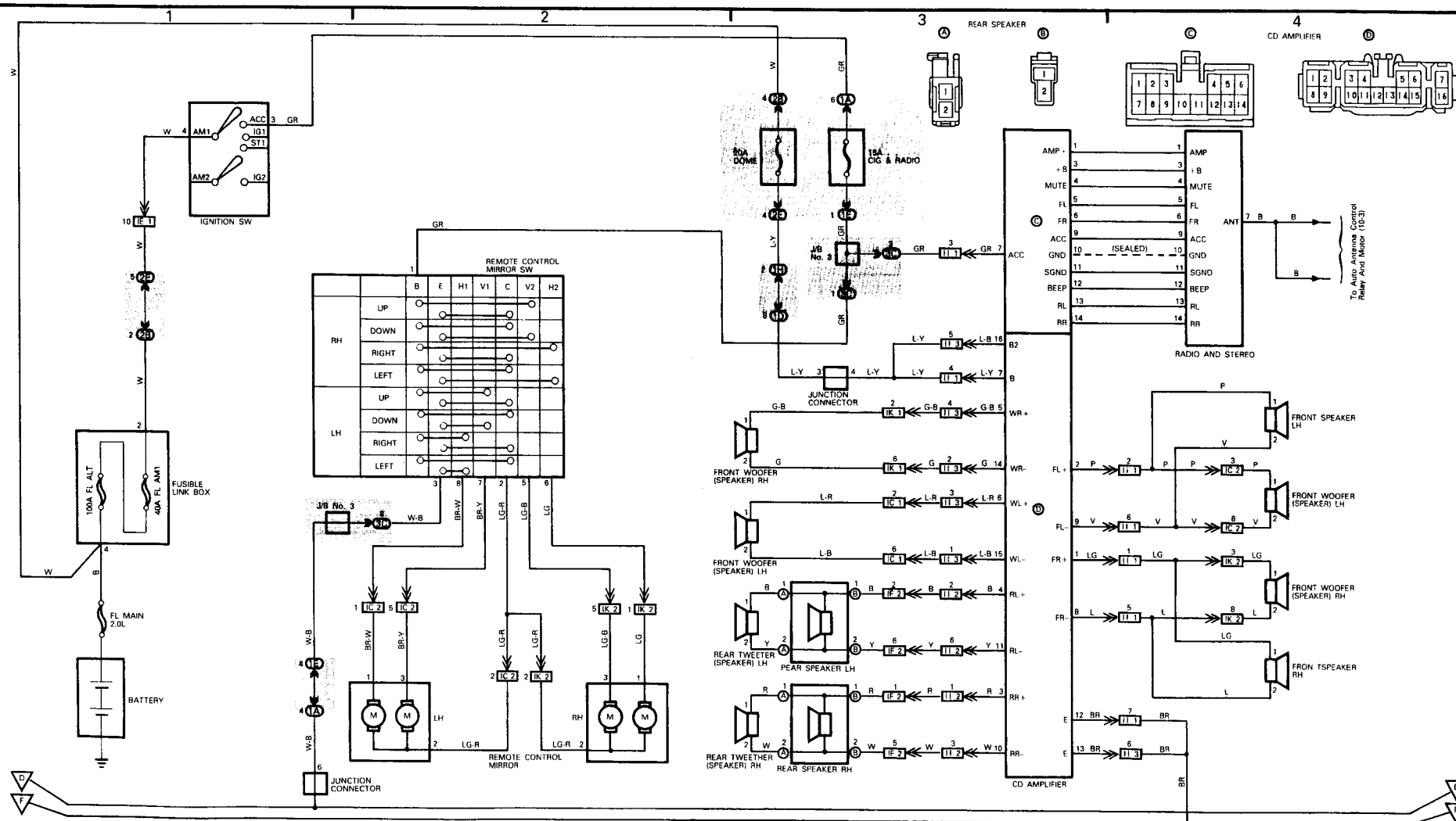
Power Source



Remote Control Mirrors



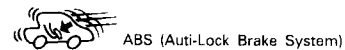
Radio and Player (w/ CD player)



Ground points

△ D = Located on left kick panel

△ F = Located on instrument panel brace RH



 – Located on front side of right fender

 = Located under left rear center pillar

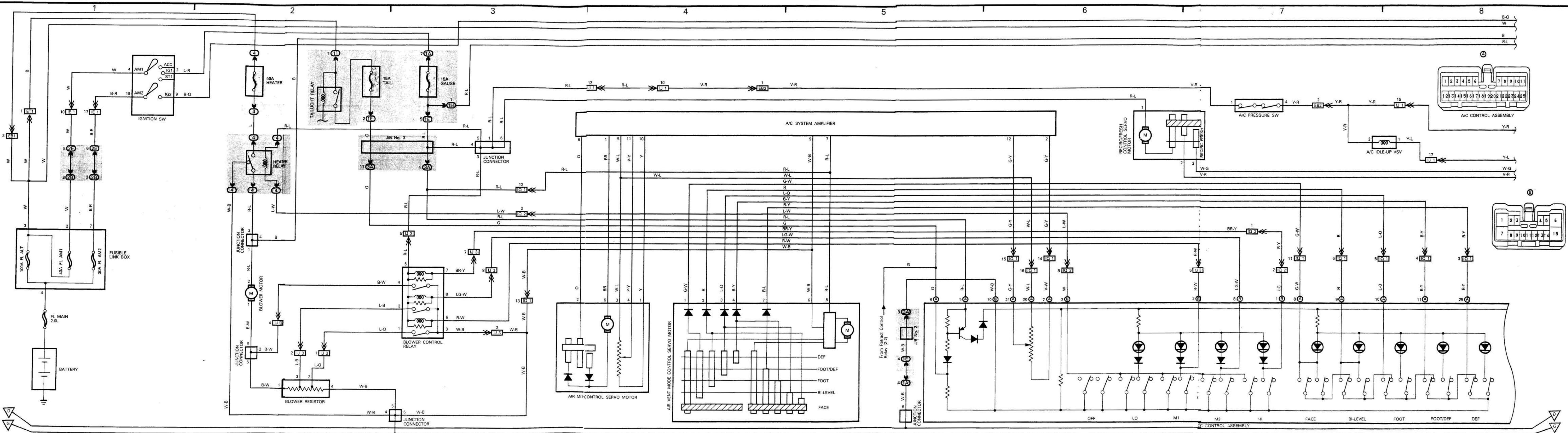




Power Source



Air Conditioner, Cooler and Heater (Manual)



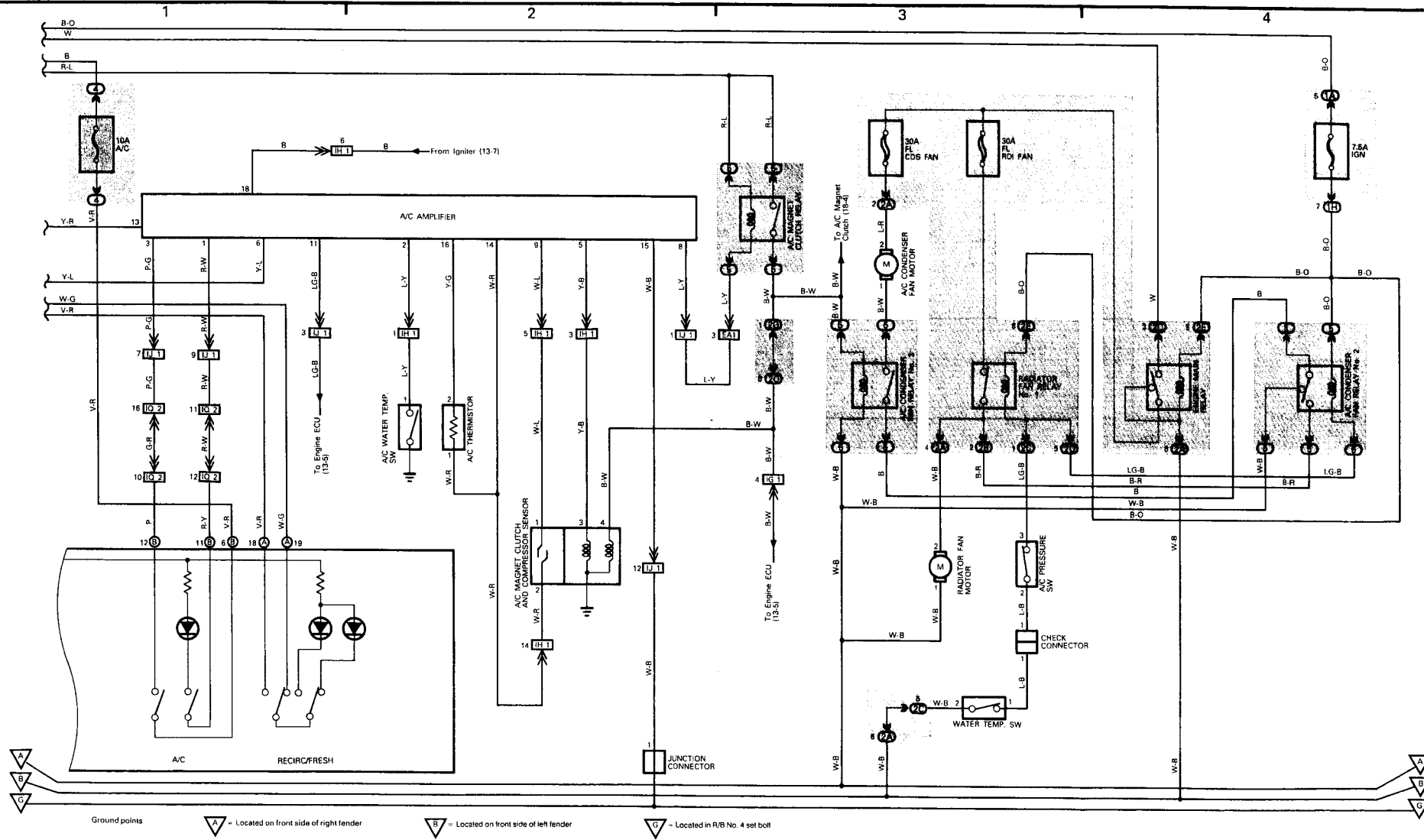
Ground points
D = Located on left kick panel
G = Located in R/B No. 4 set bolt



Air Conditioner, Cooler and Heater (Manual)



Radiator Fan and Condenser Fan

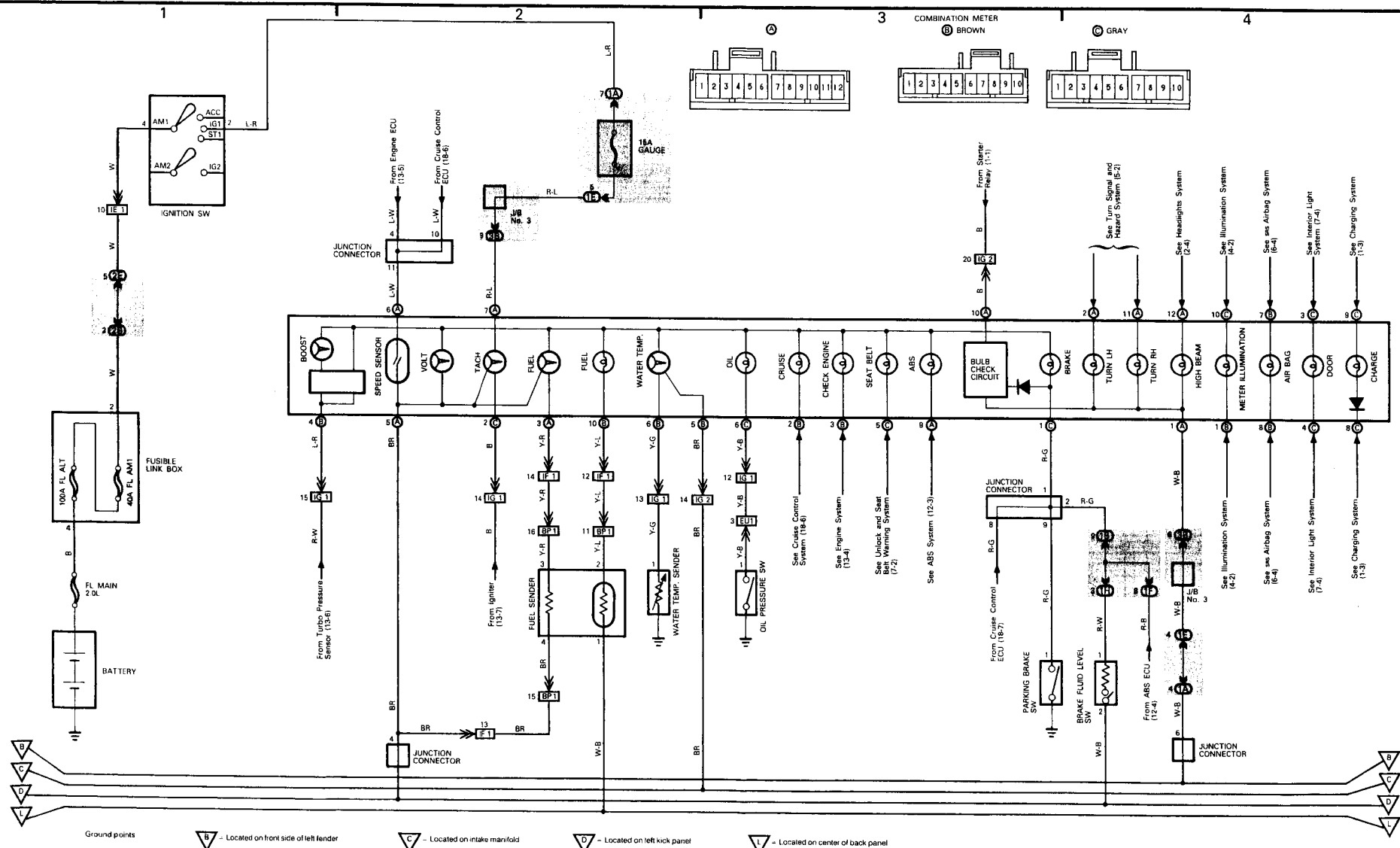


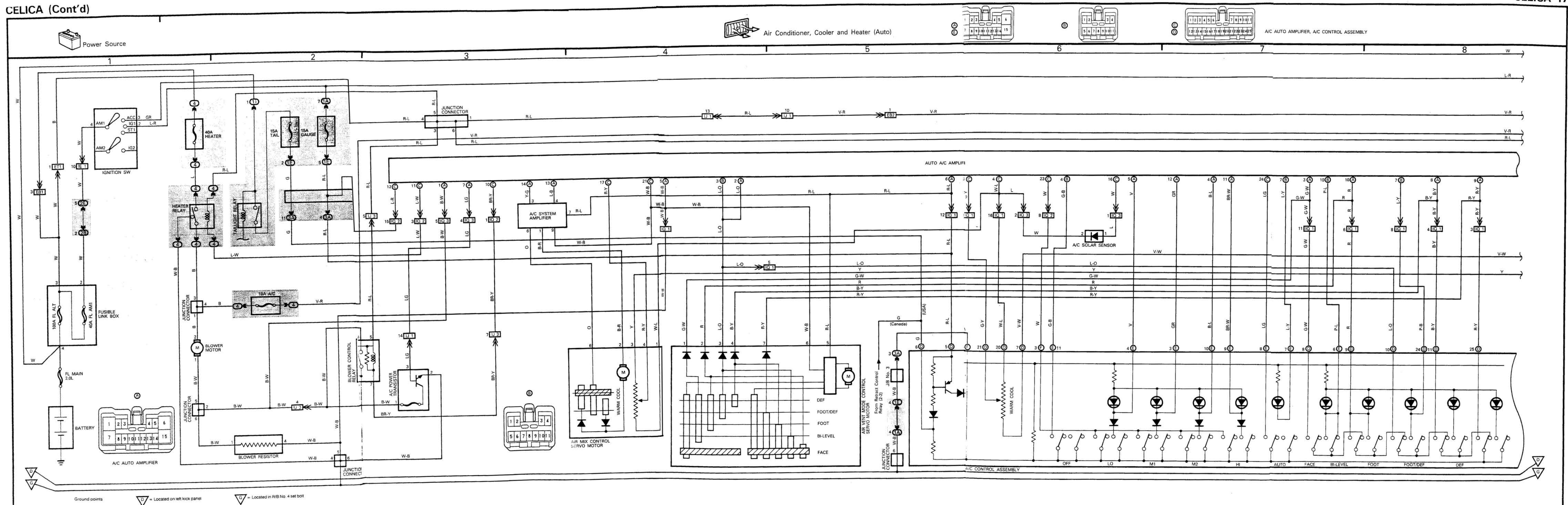


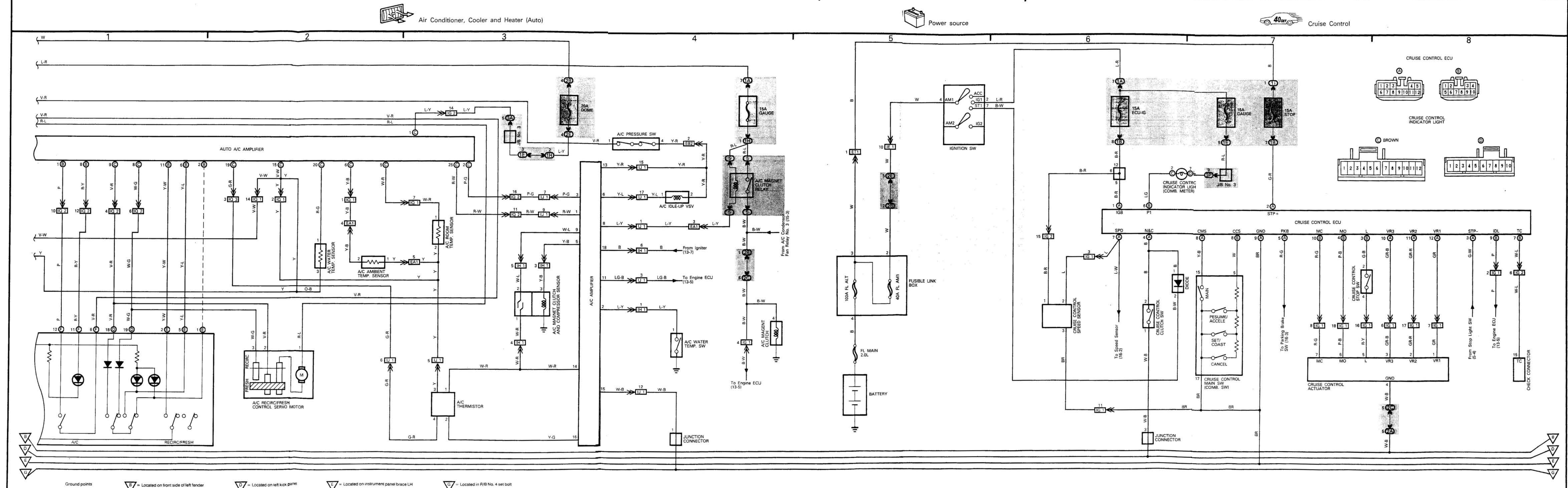
Power Source



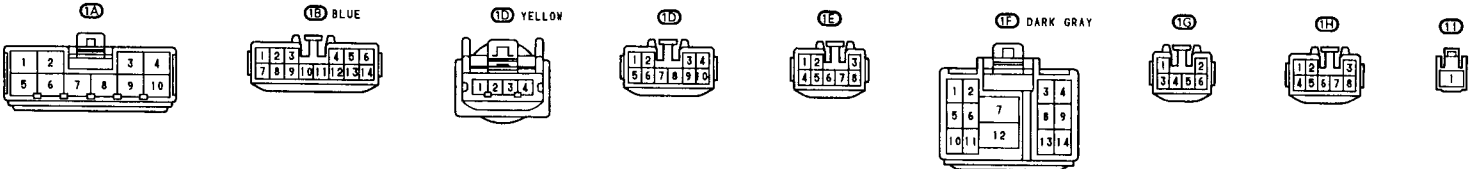
Combination Meter



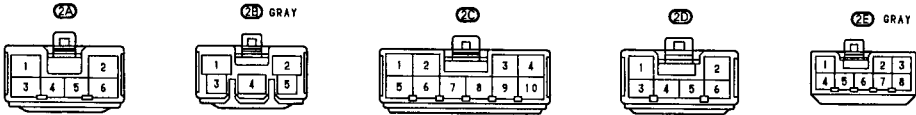




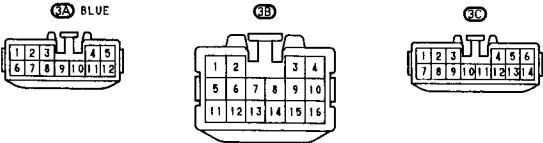
Junction Block and Wire Harness Connector
J/B No. 1



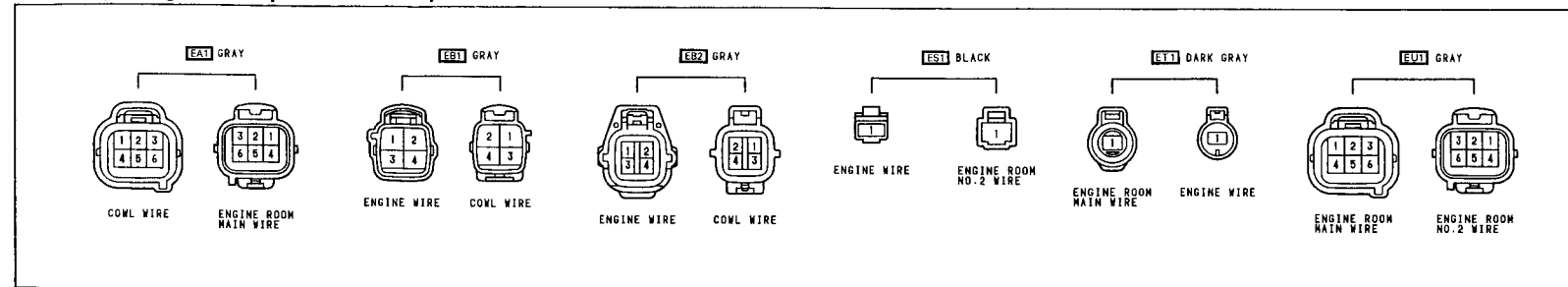
J/B No. 2



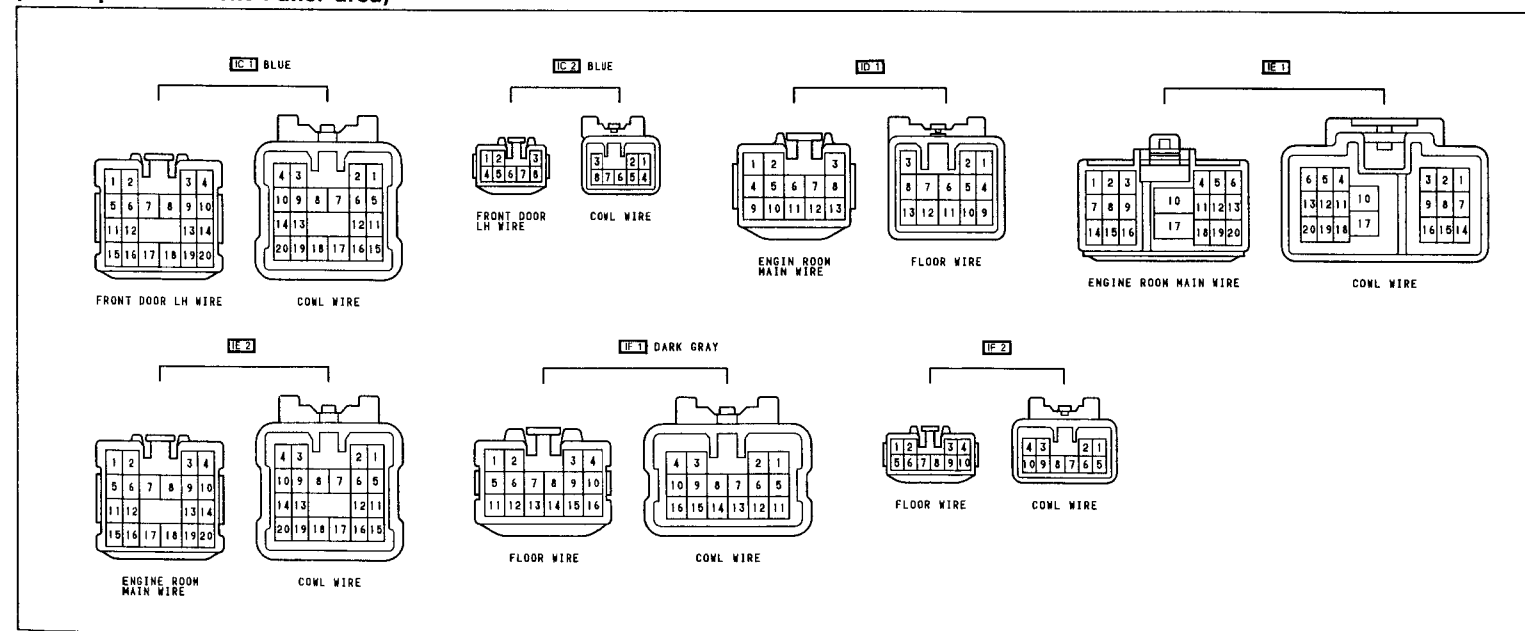
J/B No. 3



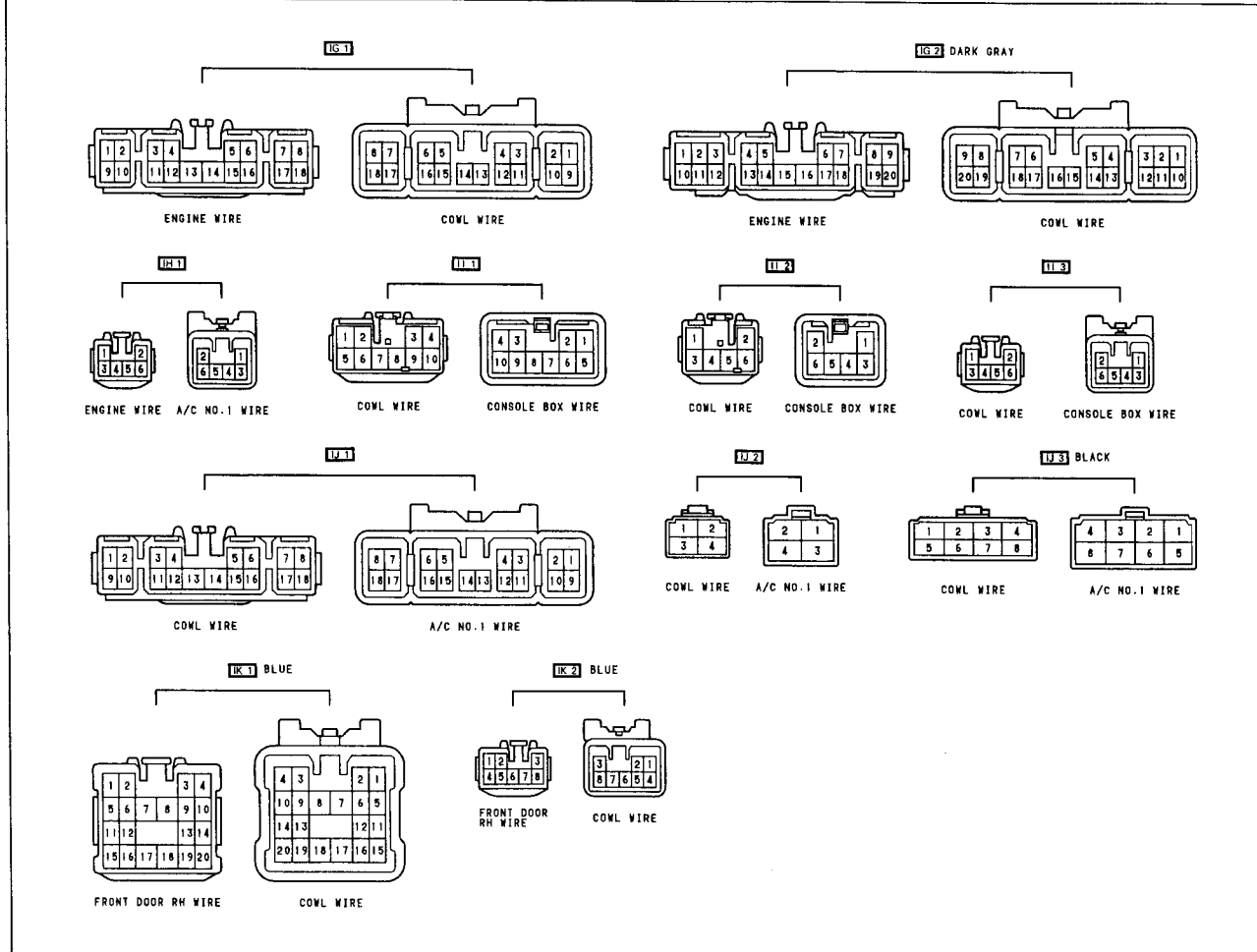
Connector Joining Wire Harness and Wire Harness (E Group: Engine Compartment area)



(I Group: Instrument Panel area)



Connector Joining Wire Harness and Wire Harness (I Group: Instrument Panel area)



Connector Joining Wire Harness and Wire Harness (B Group : Body and Surrounding area)

